

$$\Rightarrow \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Q.E.D.

Similarly, it can be proved that

$$\cot(A + B) = \frac{\cot A \cot B - 1}{\cot B + \cot A} \text{ and, } \cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

7.3 MORE USEFUL RESULTS

THEOREM Prove that:

- (i) $\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$
- (ii) $\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$
- (iii) $\sin(A + B + C) = \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C$
- (iv) $\cos(A + B + C) = \cos A \cos B \cos C - \cos A \sin B \sin C - \sin A \cos B \sin C - \sin A \sin B \cos C$
- (v) $\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$

PROOF (i) $\sin(A + B) \sin(A - B)$

$$\begin{aligned} &= (\sin A \cos B + \cos A \sin B) (\sin A \cos B - \cos A \sin B) \\ &= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B \\ &= \sin^2 A (1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B \\ &= \sin^2 A - \sin^2 A \sin^2 B - \sin^2 B + \sin^2 A \sin^2 B \\ &= \sin^2 A - \sin^2 B = (1 - \cos^2 A) - (1 - \cos^2 B) = \cos^2 B - \cos^2 A \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad &\cos(A + B) \cos(A - B) \\ &= (\cos A \cos B - \sin A \sin B) (\cos A \cos B + \sin A \sin B) \\ &= \cos^2 A \cos^2 B - \sin^2 A \sin^2 B = \cos^2 A (1 - \sin^2 B) - (1 - \cos^2 A) \sin^2 B \\ &= \cos^2 A - \sin^2 B = (1 - \sin^2 A) - (1 - \cos^2 B) = \cos^2 B - \sin^2 A \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad &\sin(A + B + C) \\ &= \sin((A + B) + C) = \sin(A + B) \cos C + \cos(A + B) \sin C \\ &= (\sin A \cos B + \cos A \sin B) \cos C + (\cos A \cos B - \sin A \sin B) \sin C \\ &= \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad &\cos(A + B + C) \\ &= \cos((A + B) + C) = \cos(A + B) \cos C - \sin(A + B) \sin C \\ &= (\cos A \cos B - \sin A \sin B) \cos C - (\sin A \cos B + \cos A \sin B) \sin C \\ &= \cos A \cos B \cos C - \sin A \sin B \cos C - \sin A \cos B \sin C - \cos A \sin B \sin C \end{aligned}$$

$$\text{(v)} \quad \tan(A + B + C) = \tan((A + B) + C)$$

$$\Rightarrow \tan(A + B + C) = \frac{\tan(A + B) + \tan C}{1 - \tan(A + B) \tan C}$$

$$\Rightarrow \tan(A + B + C) = \frac{\left(\frac{\tan A + \tan B}{1 - \tan A \tan B} \right) + \tan C}{1 - \left(\frac{\tan A + \tan B}{1 - \tan A \tan B} \right) \tan C}$$

$$\Rightarrow \tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

Q.E.D.

ILLUSTRATIVE EXAMPLES**LEVEL-1**

Type I ON FINDING THE VALUES OF $\sin(A \pm B)$, $\cos(A \pm B)$ AND $\tan(A \pm B)$ WHEN ONE OF THE TRIGONOMETRIC RATIOS OF EACH OF ANGLES A AND B IS GIVEN

EXAMPLE 1 If $\sin A = \frac{3}{5}$ and $\cos B = \frac{9}{41}$, $0 < A < \frac{\pi}{2}$, $0 < B < \frac{\pi}{2}$, find the values of the following:

- (i) $\sin(A - B)$ (ii) $\sin(A + B)$ (iii) $\cos(A - B)$ (iv) $\cos(A + B)$

SOLUTION We have,

$$\sin A = \frac{3}{5} \text{ and } \cos B = \frac{9}{41}$$

$$\therefore \cos A = \sqrt{1 - \sin^2 A} \text{ and } \sin B = \sqrt{1 - \cos^2 B}$$

$$\Rightarrow \cos A = \sqrt{1 - \frac{9}{25}} = \frac{4}{5} \text{ and } \sin B = \sqrt{1 - \frac{81}{1681}} = \frac{40}{41}$$

$$(i) \sin(A - B) = \sin A \cos B - \cos A \sin B = \frac{3}{5} \times \frac{9}{41} - \frac{4}{5} \times \frac{40}{41} = -\frac{133}{205}$$

$$(ii) \sin(A + B) = \sin A \cos B + \cos A \sin B = \frac{3}{5} \times \frac{9}{41} + \frac{4}{5} \times \frac{40}{41} = \frac{187}{205}$$

$$(iii) \cos(A - B) = \cos A \cos B + \sin A \sin B = \frac{4}{5} \times \frac{9}{41} + \frac{3}{5} \times \frac{40}{41} = \frac{156}{205}$$

$$(iv) \cos(A + B) = \cos A \cos B - \sin A \sin B = \frac{4}{5} \times \frac{9}{41} - \frac{3}{5} \times \frac{40}{41} = -\frac{84}{205}$$

EXAMPLE 2 If $\sin A = \frac{3}{5}$, $0 < A < \frac{\pi}{2}$ and $\cos B = \frac{-12}{13}$, $\pi < B < \frac{3\pi}{2}$, find the following:

- (i) $\sin(A - B)$ (ii) $\cos(A + B)$ (iii) $\tan(A - B)$

SOLUTION We have, $\sin A = \frac{3}{5}$, where $0 < A < \frac{\pi}{2}$.

$$\therefore \cos A = \pm \sqrt{1 - \sin^2 A} \Rightarrow \cos A = +\sqrt{1 - \sin^2 A} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$$

In the first quadrant tangent function is positive. Therefore, $\tan A = \frac{\sin A}{\cos A} = \frac{3}{4}$

It is given that: $\cos B = \frac{-12}{13}$ and $\pi < B < \frac{3\pi}{2}$.

$$\therefore \sin B = \pm \sqrt{1 - \cos^2 B}$$

$$\Rightarrow \sin B = -\sqrt{1 - \cos^2 B}$$

[$\because$ Sine is negative in the third quadrant]

$$\Rightarrow \sin B = -\sqrt{1 - \left(\frac{-12}{13}\right)^2} = -\frac{5}{13}$$

In the III quadrant tangent function is positive. Therefore, $\tan B = \frac{\sin B}{\cos B} = \frac{5}{12}$.

Now,

$$(i) \sin(A - B) = \sin A \cos B - \cos A \sin B = \frac{3}{5} \times \frac{-12}{13} - \frac{4}{5} \times \frac{-5}{13} = \frac{-16}{65}$$

$$(ii) \cos(A + B) = \cos A \cos B - \sin A \sin B = \frac{4}{5} \times \frac{-12}{13} - \frac{3}{5} \times \frac{-5}{13} = \frac{-33}{65}$$

$$(iii) \quad \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{\frac{3}{4} - \frac{5}{12}}{1 + \frac{3}{4} \times \frac{5}{12}} = \frac{16}{63}$$

EXAMPLE 3 If $\cos A = \frac{4}{5}$, $\cos B = \frac{12}{13}$, $\frac{3\pi}{2} < A$, $B < 2\pi$, find the values of the following:

(i) $\cos(A + B)$

(ii) $\sin(A - B)$

SOLUTION Since A and B both lie in the IV quadrant, it follows that $\sin A$ and $\sin B$ are negative. Therefore,

$$\sin A = -\sqrt{1 - \cos^2 A} \Rightarrow \sin A = -\sqrt{1 - \frac{16}{25}} = -\frac{3}{5}$$

$$\text{and, } \sin B = -\sqrt{1 - \cos^2 B} \Rightarrow \sin B = -\sqrt{1 - \frac{144}{169}} = -\frac{5}{13}$$

Now,

$$(i) \quad \cos(A + B) = \cos A \cos B - \sin A \sin B = \frac{4}{5} \times \frac{12}{13} - \left(-\frac{3}{5}\right)\left(-\frac{5}{13}\right) = \frac{33}{65}$$

$$(ii) \quad \sin(A - B) = \sin A \cos B - \cos A \sin B = \frac{-3}{5} \times \frac{12}{13} - \frac{4}{5} \times \frac{-5}{13} = \frac{-16}{65}$$

EXAMPLE 4 If $\cot \alpha = \frac{1}{2}$, $\sec \beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$. Find the value of $\tan(\alpha + \beta)$. State the quadrant in which $\alpha + \beta$ terminates.

SOLUTION We have,

$$\cot \alpha = \frac{1}{2} \Rightarrow \tan \alpha = 2$$

Since β lies in the second quadrant. Therefore, $\tan \beta$ is negative. Consequently,

$$1 + \tan^2 \beta = \sec^2 \beta \Rightarrow \tan \beta = -\sqrt{\sec^2 \beta - 1} = -\sqrt{\frac{25}{9} - 1} = -\frac{4}{3}$$

$$\text{Hence, } \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{2 - \frac{4}{3}}{1 - 2 \times \frac{-4}{3}} = \frac{2}{11}$$

$$\text{Now, } \pi < \alpha < \frac{3\pi}{2} \text{ and } \frac{\pi}{2} < \beta < \pi \Rightarrow \frac{3\pi}{2} < \alpha + \beta < \frac{5\pi}{2}$$

We know that tangent function is positive in I and III quadrants.

$$\therefore \frac{3\pi}{2} < \alpha + \beta < \frac{5\pi}{2} \text{ and } \tan(\alpha + \beta) = \frac{2}{11} > 0 \Rightarrow \alpha + \beta \text{ lies in I quadrant.}$$

Type II ON FINDING THE TRIGONOMETRIC RATIOS OF ANGLES WHICH ARE MULTIPLES OF 15°

EXAMPLE 5 Find the values of the following:

(i) $\sin 75^\circ$ [NCERT] (ii) $\cos 75^\circ$ [NCERT] (iii) $\sin 15^\circ$ (iv) $\cos 15^\circ$

SOLUTION (i) $\sin 75^\circ = \sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$

$$\Rightarrow \sin 75^\circ = \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$(ii) \quad \cos 75^\circ = \cos(45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$$

$$\Rightarrow \cos 75^\circ = \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

$$\begin{aligned}
 \text{(iii)} \quad \sin 15^\circ &= \sin (45^\circ - 30^\circ) = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\
 \Rightarrow \sin 15^\circ &= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3} - 1}{2\sqrt{2}} \\
 \text{(iv)} \quad \cos 15^\circ &= \cos (45^\circ - 30^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\
 \Rightarrow \cos 15^\circ &= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3} + 1}{2\sqrt{2}}
 \end{aligned}$$

EXAMPLE 6 Find the values of the following:

$$\text{(i) } \tan 15^\circ \text{ [NCERT]} \quad \text{(ii) } \tan 75^\circ \quad \text{(iii) } \tan 105^\circ \quad \text{(iv) } \tan \frac{13\pi}{12} \quad \text{[NCERT]}$$

$$\text{SOLUTION (i) } \tan 15^\circ = \tan (45^\circ - 30^\circ) = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ} = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$\text{(ii) } \tan 75^\circ = \tan (45^\circ + 30^\circ) = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$$\text{(iii) } \tan 105^\circ = \tan (90^\circ + 15^\circ) = -\cot 15^\circ = -\frac{1}{\tan 15^\circ} = -\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \quad \text{[Using (i)]}$$

$$\text{(iv) } \tan \frac{13\pi}{12} = \tan \left(\pi + \frac{\pi}{12} \right) = \tan \frac{\pi}{12} = \tan 15^\circ = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} \quad \text{[Using (i)]}$$

EXAMPLE 7 Prove that: $\tan 75^\circ + \cot 75^\circ = 4$

SOLUTION We have,

$$\tan 75^\circ = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \quad \text{[See Ex. 6]}$$

$$\therefore \cot 75^\circ = \frac{1}{\tan 75^\circ} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$\begin{aligned}
 \text{Now, LHS} &= \tan 75^\circ + \cot 75^\circ = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} + \frac{\sqrt{3} - 1}{\sqrt{3} + 1} = \frac{(\sqrt{3} + 1)^2 + (\sqrt{3} - 1)^2}{(\sqrt{3} - 1)(\sqrt{3} + 1)} \\
 &= \frac{(4 + 2\sqrt{3}) + (4 - 2\sqrt{3})}{3 - 1} = \frac{8}{2} = 4 = \text{RHS}
 \end{aligned}$$

Type III ON THE APPLICATIONS OF THE FOLLOWING FORMULAE:

$$\text{(i) } \sin A \cos B \pm \cos A \sin B = \sin (A \pm B) \quad \text{(ii) } \cos A \cos B \pm \sin A \sin B = \cos (A \mp B)$$

EXAMPLE 8 Evaluate the following:

$$\begin{aligned}
 \text{(i) } \sin \frac{7\pi}{12} \cos \frac{\pi}{4} - \cos \frac{7\pi}{12} \sin \frac{\pi}{4} & \quad \text{(ii) } \sin \frac{\pi}{4} \cos \frac{\pi}{12} + \cos \frac{\pi}{4} \sin \frac{\pi}{12} \\
 \text{(iii) } \cos \frac{2\pi}{3} \cos \frac{\pi}{4} - \sin \frac{2\pi}{3} \sin \frac{\pi}{4} &
 \end{aligned}$$

$$\text{SOLUTION (i) } \sin \frac{7\pi}{12} \cos \frac{\pi}{4} - \cos \frac{7\pi}{12} \sin \frac{\pi}{4} = \sin \left(\frac{7\pi}{12} - \frac{\pi}{4} \right) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\text{(ii) } \sin \frac{\pi}{4} \cos \frac{\pi}{12} + \cos \frac{\pi}{4} \sin \frac{\pi}{12} = \sin \left(\frac{\pi}{4} + \frac{\pi}{12} \right) = \sin \frac{4\pi}{12} = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\begin{aligned}
 \text{(iii)} \quad \cos \frac{2\pi}{3} \cos \frac{\pi}{4} - \sin \frac{2\pi}{3} \sin \frac{\pi}{4} &= \cos \left(\frac{2\pi}{3} + \frac{\pi}{4} \right) = \cos \frac{11\pi}{12} = \cos 165^\circ \\
 &= \cos (180^\circ - 15^\circ) = -\cos 15^\circ = -\frac{\sqrt{3}+1}{2\sqrt{2}} \quad [\text{See Ex. 5 (iv)}]
 \end{aligned}$$

EXAMPLE 9 *Prove that:* $\cos\left(\frac{\pi}{4}-A\right)\cos\left(\frac{\pi}{4}-B\right)-\sin\left(\frac{\pi}{4}-A\right)\sin\left(\frac{\pi}{4}-B\right)=\sin(A+B)$ [NCERT]

SOLUTION We have,

$$\begin{aligned}
 \text{LHS} &= \cos\left(\frac{\pi}{4}-A\right)\cos\left(\frac{\pi}{4}-B\right)-\sin\left(\frac{\pi}{4}-A\right)\sin\left(\frac{\pi}{4}-B\right) \\
 &= \cos\left\{\left(\frac{\pi}{4}-A\right)+\left(\frac{\pi}{4}-B\right)\right\} = \cos\left\{\frac{\pi}{2}-(A+B)\right\} = \sin(A+B) = \text{RHS}
 \end{aligned}$$

EXAMPLE 10 *Prove that:* $\sin(n+1)A \sin(n+2)A + \cos(n+1)A \cos(n+2)A = \cos A$ [NCERT]

$$\begin{aligned}
 \text{SOLUTION} \quad \text{LHS} &= \sin(n+1)A \sin(n+2)A + \cos(n+1)A \cos(n+2)A \\
 &= \cos(n+2)A \cos(n+1)A + \sin(n+2)A \sin(n+1)A \\
 &= \cos\{(n+2)A - (n+1)A\} = \cos A = \text{RHS}
 \end{aligned}$$

Type IV ON APPLICATIONS OF THE FORMULAE:

$$\text{(i)} \quad \sin(A \pm B) = \sin A \cos B \pm \cos A \sin B \quad \text{(ii)} \quad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\text{(iii)} \quad \tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

EXAMPLE 11 *Prove that:*

$$\text{(i)} \quad \cos\left(\frac{\pi}{4}+x\right) + \cos\left(\frac{\pi}{4}-x\right) = \sqrt{2} \cos x \quad [\text{NCERT}]$$

$$\text{(ii)} \quad \cos\left(\frac{3\pi}{4}+x\right) - \cos\left(\frac{3\pi}{4}-x\right) = -\sqrt{2} \sin x \quad [\text{NCERT}]$$

SOLUTION (i) We have,

$$\begin{aligned}
 &\cos\left(\frac{\pi}{4}+x\right) + \cos\left(\frac{\pi}{4}-x\right) \\
 &= \left(\cos \frac{\pi}{4} \cos x - \sin \frac{\pi}{4} \sin x\right) + \left(\cos \frac{\pi}{4} \cos x + \sin \frac{\pi}{4} \sin x\right) \\
 &= 2 \cos \frac{\pi}{4} \cos x = 2 \times \frac{1}{\sqrt{2}} \times \cos x = \sqrt{2} \cos x
 \end{aligned}$$

(ii) We have,

$$\begin{aligned}
 &\cos\left(\frac{3\pi}{4}+x\right) - \cos\left(\frac{3\pi}{4}-x\right) = -\sqrt{2} \sin x \\
 &= \left(\cos \frac{3\pi}{4} \cos x - \sin \frac{3\pi}{4} \sin x\right) + \left(\cos \frac{3\pi}{4} \cos x + \sin \frac{3\pi}{4} \sin x\right) \\
 &= \left(-\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x\right) - \left(-\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x\right) \\
 &= -\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x - \left(-\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x\right) \\
 &= -\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x = -\sqrt{2} \sin x
 \end{aligned}$$

EXAMPLE 12 *Prove that:* $\frac{\sin(x+y)}{\sin(x-y)} = \frac{\tan x + \tan y}{\tan x - \tan y}$ [NCERT]

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{\sin(x+y)}{\sin(x-y)} = \frac{\sin x \cos y + \cos x \sin y}{\sin x \cos y - \cos x \sin y} \\ &= \frac{\sin x \cos y + \cos x \sin y}{\sin x \cos y - \cos x \sin y} \quad [\text{Dividing the numerator and denominator by } \cos x \cos y] \\ &= \frac{\cos x \cos y}{\sin x \cos y - \cos x \sin y} \\ &= \frac{\tan x + \tan y}{\tan x - \tan y} = \text{RHS} \end{aligned}$$

EXAMPLE 13 ✓ Prove that: $\frac{\sin(B-C)}{\cos B \cos C} + \frac{\sin(C-A)}{\cos C \cos A} + \frac{\sin(A-B)}{\cos A \cos B} = 0$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{\sin(B-C)}{\cos B \cos C} + \frac{\sin(C-A)}{\cos C \cos A} + \frac{\sin(A-B)}{\cos A \cos B} \\ &= \frac{\sin B \cos C - \cos B \sin C}{\cos B \cos C} + \frac{\sin C \cos A - \cos C \sin A}{\cos C \cos A} + \frac{\sin A \cos B - \cos A \sin B}{\cos A \cos B} \\ &= \frac{\sin B \cos C}{\cos B \cos C} - \frac{\cos B \sin C}{\cos B \cos C} + \frac{\sin C \cos A}{\cos C \cos A} - \frac{\cos C \sin A}{\cos C \cos A} + \frac{\sin A \cos B}{\cos A \cos B} - \frac{\cos A \sin B}{\cos A \cos B} \\ &= \tan B - \tan C + \tan C - \tan A + \tan A - \tan B = 0 = \text{RHS} \end{aligned}$$

EXAMPLE 14 ✓ Prove that: $\frac{\tan\left(\frac{\pi}{4} + x\right)}{\tan\left(\frac{\pi}{4} - x\right)} = \left(\frac{1 + \tan x}{1 - \tan x}\right)^2$ [NCERT]

$$\begin{aligned} \text{SOLUTION} \quad \text{LHS} &= \frac{\tan\left(\frac{\pi}{4} + x\right)}{\tan\left(\frac{\pi}{4} - x\right)} = \frac{\tan \frac{\pi}{4} + \tan x}{1 - \tan \frac{\pi}{4} \tan x} \times \frac{1 + \tan \frac{\pi}{4} \tan x}{\tan \frac{\pi}{4} - \tan x} \\ &= \frac{1 + \tan x}{1 - \tan x} \times \frac{1 + \tan x}{1 - \tan x} = \left(\frac{1 + \tan x}{1 - \tan x}\right)^2 = \text{RHS} \end{aligned}$$

EXAMPLE 15 ✓ If $\tan A - \tan B = x$ and $\cot B - \cot A = y$, prove that $\cot(A-B) = \frac{1}{x} + \frac{1}{y}$

[NCERT EXEMPAR]

SOLUTION We have, $\tan A - \tan B = x$ and $\cot B - \cot A = y$

Now,

$$\Rightarrow \frac{\cot B - \cot A}{\tan B - \tan A} = y \Rightarrow \frac{\frac{1}{\tan B} - \frac{1}{\tan A}}{\tan A - \tan B} = y \Rightarrow \frac{\frac{\tan A - \tan B}{\tan A \tan B}}{\tan A - \tan B} = y \Rightarrow \frac{x}{\tan A \tan B} = y \Rightarrow \tan A \tan B = \frac{x}{y}$$

$$\therefore \cot(A-B) = \frac{1}{\tan(A-B)} = \frac{1 + \tan A \tan B}{\tan A - \tan B} = \frac{1 + \frac{x}{y}}{x} = \frac{x+y}{xy} = \frac{1}{x} + \frac{1}{y}$$

EXAMPLE 16 If $\tan \alpha = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan \beta = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and $\tan \gamma = \sqrt{x^{-3} + x^{-2} + x^{-1}}$,

prove that $\alpha + \beta = \gamma$.

SOLUTION We have,

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{\frac{1}{\sqrt{x(x^2+x+1)}} + \frac{\sqrt{x}}{\sqrt{x^2+x+1}}}{1 - \frac{1}{\sqrt{x(x^2+x+1)}} \times \frac{\sqrt{x}}{\sqrt{x^2+x+1}}} = \frac{(x+1)\sqrt{x(x^2+x+1)}}{x(x^2+x+1) - x}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{(x+1)\sqrt{x(x^2+x+1)}}{x^2(x+1)} = \frac{\sqrt{x(x^2+x+1)}}{x^2} = \sqrt{\frac{x(x^2+x+1)}{x^4}}$$

$$\Rightarrow \tan(\alpha + \beta) = \sqrt{x^{-3} + x^{-2} + x^{-1}} = \tan \gamma$$

$$\Rightarrow \alpha + \beta = \gamma.$$

EXAMPLE 17 If α and β are acute angles such that $\tan \alpha = \frac{m}{m+1}$ and $\tan \beta = \frac{1}{2m+1}$, prove that

$$\alpha + \beta = \frac{\pi}{4}.$$

SOLUTION We have,

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{\frac{m}{m+1} + \frac{1}{2m+1}}{1 - \frac{m}{m+1} \times \frac{1}{2m+1}} = \frac{\frac{2m^2 + m + m + 1}{(m+1)(2m+1)}}{\frac{2m^2 + 3m + 1 - m}{(m+1)(2m+1)}} = \frac{2m^2 + 2m + 1}{2m^2 + 3m + 1 - m} = 1 = \tan \frac{\pi}{4}$$

$$\Rightarrow \alpha + \beta = \frac{\pi}{4}$$

EXAMPLE 18 If $A + B = \frac{\pi}{4}$, prove that:

$$(i) (1 + \tan A)(1 + \tan B) = 2$$

$$(ii) (\cot A - 1)(\cot B - 1) = 2$$

SOLUTION (i) We have,

$$A + B = \frac{\pi}{4}$$

$$\Rightarrow \tan(A + B) = \tan \frac{\pi}{4}$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$

$$\Rightarrow \tan A + \tan B = 1 - \tan A \tan B$$

$$\Rightarrow \tan A + \tan B + \tan A \tan B = 1$$

$$\Rightarrow 1 + \tan A + \tan B + \tan A \tan B = 2$$

$$\Rightarrow (1 + \tan A) + \tan B(1 + \tan A) = 2 \Rightarrow (1 + \tan A)(1 + \tan B) = 2$$

(ii) We have,

$$A + B = \frac{\pi}{4}$$

$$\Rightarrow \tan(A + B) = \tan \frac{\pi}{4}$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$

$$\Rightarrow \tan A + \tan B = 1 - \tan A \tan B$$

$$\begin{aligned} \Rightarrow \tan A + \tan B + \tan A \tan B &= 1 \\ \Rightarrow \frac{\tan A + \tan B + \tan A \tan B}{\tan A \tan B} &= \frac{1}{\tan A \tan B} \\ \Rightarrow \cot B + \cot A + 1 &= \cot A \cot B \\ \Rightarrow \cot A \cot B - \cot A - \cot B &= 1 \\ \Rightarrow \cot A \cot B - \cot A - \cot B + 1 &= 2 \\ \Rightarrow \cot A (\cot B - 1) - (\cot B - 1) &= 2 \\ \Rightarrow (\cot A - 1) (\cot B - 1) &= 2 \end{aligned}$$

EXAMPLE 19 If $\tan \beta = \frac{n \sin \alpha \cos \alpha}{1 - n \sin^2 \alpha}$, show that $\tan (\alpha - \beta) = (1 - n) \tan \alpha$.

SOLUTION We have,

$$\begin{aligned} \tan (\alpha - \beta) &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\ \Rightarrow \tan (\alpha - \beta) &= \frac{\frac{\sin \alpha}{\cos \alpha} - \frac{n \sin \alpha \cos \alpha}{1 - n \sin^2 \alpha}}{1 + \frac{\sin \alpha}{\cos \alpha} \times \frac{n \sin \alpha \cos \alpha}{1 - n \sin^2 \alpha}} \quad \left[\because \tan \beta = \frac{n \sin \alpha \cos \alpha}{1 - n \sin^2 \alpha} \right] \\ \Rightarrow \tan (\alpha - \beta) &= \frac{\sin \alpha - n \sin^3 \alpha - n \sin \alpha \cos^2 \alpha}{\cos \alpha (1 - n \sin^2 \alpha) + n \sin^2 \alpha \cos \alpha} \quad [\text{On taking LCM}] \\ \Rightarrow \tan (\alpha - \beta) &= \frac{\sin \alpha - n \sin^3 \alpha - n \sin \alpha (1 - \sin^2 \alpha)}{\cos \alpha - n \sin^2 \alpha \cos \alpha + n \sin^2 \alpha \cos \alpha} \\ \Rightarrow \tan (\alpha - \beta) &= \frac{\sin \alpha - n \sin \alpha}{\cos \alpha} = \frac{\sin \alpha}{\cos \alpha} (1 - n) = (1 - n) \tan \alpha \end{aligned}$$

EXAMPLE 20 Prove that:

$$(i) \tan 3A \tan 2A \tan A = \tan 3A - \tan 2A - \tan A$$

[NCERT]

$$(ii) \cot A \cot 2A - \cot 2A \cot 3A - \cot 3A \cot A = 1$$

SOLUTION (i) Clearly,

$$3A = 2A + A$$

$$\Rightarrow \tan 3A = \tan (2A + A)$$

$$\Rightarrow \tan 3A = \frac{\tan 2A + \tan A}{1 - \tan 2A \tan A}$$

$$\Rightarrow \tan 3A (1 - \tan 2A \tan A) = \tan 2A + \tan A$$

$$\Rightarrow \tan 3A - \tan 3A \tan 2A \tan A = \tan 2A + \tan A$$

$$\Rightarrow \tan 3A - \tan 2A - \tan A = \tan 3A \tan 2A \tan A$$

(ii) Dividing both sides by $\tan A \tan 2A \tan 3A$, we get

$$\frac{\tan 3A \tan 2A \tan A}{\tan 3A \tan 2A \tan A} = \frac{\tan 3A - \tan 2A - \tan A}{\tan 3A \tan 2A \tan A}$$

$$\Rightarrow 1 = \frac{1}{\tan 2A \tan A} - \frac{1}{\tan 3A \tan A} - \frac{1}{\tan 3A \tan 2A}$$

$$\Rightarrow 1 = \cot A \cot 2A - \cot 3A \cot A - \cot 3A \cot 2A$$

Type V ON THE APPLICATIONS OF THE FORMULAE:

$$(i) \sin(A+B) \sin(A-B) = \sin^2 A - \sin^2 B \quad (ii) \cos(A+B) \cos(A-B) = \cos^2 A - \sin^2 B$$

EXAMPLE 21 ✓ Prove that: $\frac{\tan(A+B)}{\cot(A-B)} = \frac{\sin^2 A - \sin^2 B}{\cos^2 A - \sin^2 B}$

SOLUTION LHS = $\frac{\tan(A+B)}{\cot(A-B)} = \frac{\sin(A+B)}{\cos(A+B)} \cdot \frac{\sin(A-B)}{\cos(A-B)} = \frac{\sin^2 A - \sin^2 B}{\cos^2 A - \sin^2 B} = \text{RHS}$

EXAMPLE 22 ✓ Prove that : $\sin^2 6x - \sin^2 4x = \sin 2x \sin 10x$

[NCERT]

SOLUTION We know that $\sin^2 A - \sin^2 B = \sin(A+B) \sin(A-B)$

$$\therefore \sin^2 6x - \sin^2 4x = \sin(6x+4x) \sin(6x-4x) = \sin 10x \sin 2x$$

EXAMPLE 23 ✓ Prove that : $\cos^2 2x - \cos^2 6x = \sin 4x \sin 8x$

[NCERT]

SOLUTION We have,

$$\begin{aligned} \therefore \text{LHS} &= \cos^2 2x - \cos^2 6x \\ &= (1 - \sin^2 2x) - (1 - \sin^2 6x) \\ &= \sin^2 6x - \sin^2 2x = \sin(6x+2x) \sin(6x-2x) = \sin 8x \sin 4x = \text{RHS} \end{aligned}$$

EXAMPLE 24 ✓ Prove that $\frac{\cos^2 33^\circ - \cos^2 57^\circ}{\sin^2 \frac{21^\circ}{2} - \sin^2 \frac{69^\circ}{2}} = -\sqrt{2}$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{\cos^2 33^\circ - \cos^2 57^\circ}{\sin^2 \frac{21^\circ}{2} - \sin^2 \frac{69^\circ}{2}} \\ &= \frac{(1 - \sin^2 33^\circ) - (1 - \sin^2 57^\circ)}{\sin^2 \frac{21^\circ}{2} - \sin^2 \frac{69^\circ}{2}} \\ &= \frac{\sin^2 57^\circ - \sin^2 33^\circ}{\sin^2 \frac{21^\circ}{2} - \sin^2 \frac{69^\circ}{2}} \\ &= \frac{\sin(57^\circ + 33^\circ) \sin(57^\circ - 33^\circ)}{\sin\left(\frac{21^\circ}{2} + \frac{69^\circ}{2}\right) \sin\left(\frac{21^\circ}{2} - \frac{69^\circ}{2}\right)} \\ &= \frac{\sin 90^\circ \sin 24^\circ}{\sin 45^\circ \sin(-24^\circ)} = \frac{\sin 24^\circ}{-\frac{1}{\sqrt{2}} \sin 24^\circ} = -\sqrt{2} = \text{RHS} \end{aligned}$$

EXAMPLE 25 ✓ Prove that: $\sin^2\left(\frac{\pi}{8} + \frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{A}{2}\right) = \frac{1}{\sqrt{2}} \sin A$

SOLUTION Using $\sin^2 A - \sin^2 B = \sin(A+B) \sin(A-B)$, we obtain

$$\begin{aligned} \text{LHS} &= \sin^2\left(\frac{\pi}{8} + \frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{A}{2}\right) \\ &= \sin\left\{\left(\frac{\pi}{8} + \frac{A}{2}\right) + \left(\frac{\pi}{8} - \frac{A}{2}\right)\right\} \sin\left\{\left(\frac{\pi}{8} + \frac{A}{2}\right) - \left(\frac{\pi}{8} - \frac{A}{2}\right)\right\} \\ &= \sin \frac{\pi}{4} \sin A = \frac{1}{\sqrt{2}} \sin A = \text{RHS} \end{aligned}$$

EXAMPLE 26 ✓ Prove that: $\cos 2\alpha \cos 2\beta + \sin^2(\alpha - \beta) - \sin^2(\alpha + \beta) = \cos 2(\alpha + \beta)$

SOLUTION We have,

$$\text{LHS} = \cos 2\alpha \cos 2\beta + \sin^2(\alpha - \beta) - \sin^2(\alpha + \beta)$$

$$\begin{aligned}
 &= \cos 2\alpha \cos 2\beta + \sin(\alpha - \beta + \alpha + \beta) \sin(\alpha - \beta - \alpha - \beta) \quad \left[\begin{array}{l} \because \sin^2 A - \sin^2 B \\ = \sin(A+B) \sin(A-B) \end{array} \right] \\
 &= \cos 2\alpha \cos 2\beta + \sin 2\alpha \sin(-2\beta) \\
 &= \cos 2\alpha \cos 2\beta - \sin 2\alpha \sin 2\beta = \cos(2\alpha + 2\beta)
 \end{aligned}$$

EXAMPLE 27 Prove that: $\sin^2 A = \cos^2(A - B) + \cos^2 B - 2 \cos(A - B) \cos A \cos B$.

SOLUTION We have,

$$\begin{aligned}
 \text{RHS} &= \cos^2(A - B) + \cos^2 B - 2 \cos(A - B) \cos A \cos B \\
 &= \cos^2 B + \cos^2(A - B) - 2 \cos(A - B) \cos A \cos B \\
 &= \cos^2 B + \cos(A - B) \{ \cos(A - B) - 2 \cos A \cos B \} \\
 &= \cos^2 B + \cos(A - B) \{ \cos A \cos B + \sin A \sin B - 2 \cos A \cos B \} \\
 &= \cos^2 B + \cos(A - B) \{ \sin A \sin B - \cos A \cos B \} \\
 &= \cos^2 B - \cos(A - B) (\cos A \cos B - \sin A \sin B) \\
 &= \cos^2 B - \cos(A - B) \cos(A + B) \\
 &= \cos^2 B - (\cos^2 A - \sin^2 B) \\
 &= \cos^2 B + \sin^2 B - \cos^2 A = 1 - \cos^2 A = \sin^2 A = \text{LHS}
 \end{aligned}$$

LEVEL-2

EXAMPLE 28 If $3 \tan A \tan B = 1$, prove that $2 \cos(A + B) = \cos(A - B)$.

SOLUTION We have,

$$\begin{aligned}
 &3 \tan A \tan B = 1 \\
 \Rightarrow &\frac{3 \sin A \sin B}{\cos A \cos B} = 1 \\
 \Rightarrow &\frac{\cos A \cos B}{\sin A \sin B} = \frac{3}{1} \\
 \Rightarrow &\frac{\cos A \cos B + \sin A \sin B}{\cos A \cos B - \sin A \sin B} = \frac{3+1}{3-1} \quad [\text{Applying componendo-dividendo}] \\
 \Rightarrow &\frac{\cos(A - B)}{\cos(A + B)} = 2 \Rightarrow 2 \cos(A + B) = \cos(A - B)
 \end{aligned}$$

EXAMPLE 29 If $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -\frac{3}{2}$, prove that

$$\cos \alpha + \cos \beta + \cos \gamma = \sin \alpha + \sin \beta + \sin \gamma = 0$$

SOLUTION We have,

$$\begin{aligned}
 &\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -\frac{3}{2} \\
 \Rightarrow &2 \cos \alpha \cos \beta + 2 \sin \alpha \sin \beta + 2 \cos \beta \cos \gamma + 2 \sin \beta \sin \gamma \\
 &\quad + 2 \cos \gamma \cos \alpha + 2 \sin \gamma \sin \alpha = -3 \\
 \Rightarrow &(2 \cos \alpha \cos \beta + 2 \cos \beta \cos \gamma + 2 \cos \gamma \cos \alpha) + (2 \sin \alpha \sin \beta + 2 \sin \beta \sin \gamma \\
 &\quad + 2 \sin \gamma \sin \alpha) + 3 = 0 \\
 \Rightarrow &(2 \cos \alpha \cos \beta + 2 \cos \beta \cos \gamma + 2 \cos \gamma \cos \alpha) + (2 \sin \alpha \sin \beta + 2 \sin \beta \sin \gamma \\
 &\quad + 2 \sin \gamma \sin \alpha + (\cos^2 \alpha + \sin^2 \alpha) + (\cos^2 \beta + \sin^2 \beta) + (\cos^2 \gamma + \sin^2 \gamma)) = 0 \\
 \Rightarrow &(\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + 2 \cos \alpha \cos \beta + 2 \cos \beta \cos \gamma + 2 \cos \gamma \cos \alpha) \\
 &\quad + (\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + 2 \sin \alpha \sin \beta + 2 \sin \beta \sin \gamma + 2 \sin \gamma \sin \alpha) = 0
 \end{aligned}$$

$$\Rightarrow (\cos \alpha + \cos \beta + \cos \gamma)^2 + (\sin \alpha + \sin \beta + \sin \gamma)^2 = 0$$

$$\Rightarrow \cos \alpha + \cos \beta + \cos \gamma = 0 \text{ and } \sin \alpha + \sin \beta + \sin \gamma = 0$$

EXAMPLE 30 If $\sin B = 3 \sin (2A + B)$, prove that $2 \tan A + \tan (A + B) = 0$.

SOLUTION We have,

$$\sin B = 3 \sin (2A + B)$$

$$\Rightarrow \frac{\sin (2A + B)}{\sin B} = \frac{1}{3}$$

$$\Rightarrow \frac{\sin \{(A + B) + A\}}{\sin \{(A + B) - A\}} = \frac{1}{3}$$

$$\Rightarrow \frac{\sin [A + B + A] + \sin [(A + B) - A]}{\sin [(A + B) + A] - \sin [(A + B) - A]} = \frac{1 + 3}{1 - 3} \quad [\text{Using componendo-dividendo}]$$

$$\Rightarrow \frac{\{\sin (A + B) \cos A + \cos (A + B) \sin A\} + \{\sin (A + B) \cos A - \cos (A + B) \sin A\}}{\{\sin (A + B) \cos A + \cos (A + B) \sin A\} - \{\sin (A + B) \cos A - \cos (A + B) \sin A\}} = \frac{1 + 3}{1 - 3}$$

$$\Rightarrow \frac{2 \sin (A + B) \cos A}{2 \cos (A + B) \sin A} = -2$$

$$\Rightarrow \tan (A + B) \cot A = -2 \Rightarrow \tan (A + B) = -2 \tan A \Rightarrow 2 \tan A + \tan (A + B) = 0$$

EXAMPLE 31 If $2 \tan \beta + \cot \beta = \tan \alpha$, prove that $\cot \beta = 2 \tan (\alpha - \beta)$.

SOLUTION Clearly,

$$\begin{aligned} 2 \tan (\alpha - \beta) &= 2 \left\{ \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \right\} \\ &= 2 \left\{ \frac{2 \tan \beta + \cot \beta - \tan \beta}{1 + (2 \tan \beta + \cot \beta) \tan \beta} \right\} \quad [\text{Using : } \tan \alpha = 2 \tan \beta + \cot \beta] \\ &= 2 \left\{ \frac{\tan \beta + \cot \beta}{1 + 2 \tan^2 \beta + 1} \right\} \\ &= \frac{2 (\tan \beta + \cot \beta)}{2 + 2 \tan^2 \beta} = \frac{2 \left\{ \tan \beta + \frac{1}{\tan \beta} \right\}}{2 (1 + \tan^2 \beta)} = \frac{1}{\tan \beta} = \cot \beta \end{aligned}$$

Hence, $\cot \beta = 2 \tan (\alpha - \beta)$.

EXAMPLE 32 If $\cos (\alpha + \beta) \sin (\gamma + \delta) = \cos (\alpha - \beta) \sin (\gamma - \delta)$, prove that $\cot \alpha \cot \beta \cot \gamma = \cot \delta$.

SOLUTION We have,

$$\cos (\alpha + \beta) \sin (\gamma + \delta) = \cos (\alpha - \beta) \sin (\gamma - \delta)$$

$$\Rightarrow \frac{\cos (\alpha + \beta)}{\cos (\alpha - \beta)} = \frac{\sin (\gamma + \delta)}{\sin (\gamma - \delta)}$$

$$\Rightarrow \frac{\cos (\alpha - \beta) + \cos (\alpha + \beta)}{\cos (\alpha - \beta) - \cos (\alpha + \beta)} = \frac{\sin (\gamma + \delta) + \sin (\gamma - \delta)}{\sin (\gamma + \delta) - \sin (\gamma - \delta)} \quad [\text{Using componendo-dividendo}]$$

$$\Rightarrow \frac{2 \cos \alpha \cos \beta}{2 \sin \alpha \sin \beta} = \frac{2 \sin \gamma \cos \delta}{2 \cos \gamma \sin \delta} \Rightarrow \cot \alpha \cot \beta = \tan \gamma \cot \delta \Rightarrow \cot \alpha \cot \beta \cot \gamma = \cot \delta$$

EXAMPLE 33 Prove that: $\frac{\sin (x + \theta)}{\sin (x + \phi)} = \cos (\theta - \phi) + \cot (x + \phi) \sin (\theta - \phi)$.

SOLUTION We have,

$$\frac{\sin (x + \theta)}{\sin (x + \phi)} = \frac{\sin \{(x + \phi) + (\theta - \phi)\}}{\sin (x + \phi)}$$

$$= \frac{\sin(x + \phi) \cos(\theta - \phi) + \cos(x + \phi) \sin(\theta - \phi)}{\sin(x + \phi)}$$

$$= \cos(\theta - \phi) + \cot(x + \phi) \sin(\theta - \phi)$$

EXAMPLE 34 If $\cos(\alpha + \beta) = \frac{4}{5}$, $\sin(\alpha - \beta) = \frac{5}{13}$ and α, β lie between 0 and $\frac{\pi}{4}$, prove that

$$\tan 2\alpha = \frac{56}{33}$$

[NCERT EXEMPLAR]

SOLUTION It is given that α, β lie between 0 and $\pi/4$. Therefore, $-\pi/4 < \alpha - \beta < \pi/4$ and $0 < \alpha + \beta < \pi/2$.

So, $\cos(\alpha - \beta)$ and $\sin(\alpha + \beta)$ are positive.

$$\text{Now, } \sin(\alpha + \beta) = \sqrt{1 - \cos^2(\alpha + \beta)} \Rightarrow \sin(\alpha + \beta) = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$$

$$\text{and, } \cos(\alpha - \beta) = \sqrt{1 - \sin^2(\alpha - \beta)} \Rightarrow \cos(\alpha - \beta) = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}$$

$$\therefore \tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{3/5}{4/5} = \frac{3}{4} \text{ and, } \tan(\alpha - \beta) = \frac{\sin(\alpha - \beta)}{\cos(\alpha - \beta)} = \frac{5}{12}$$

Now,

$$\tan 2\alpha = \tan\{(\alpha + \beta) + (\alpha - \beta)\} = \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \tan(\alpha - \beta)} = \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \times \frac{5}{12}} = \frac{56}{33}$$

EXAMPLE 35 Prove that: $\tan 70^\circ = \tan 20^\circ + 2 \tan 50^\circ$.

SOLUTION We have,

$$\tan A - \tan B = \tan(A - B)(1 + \tan A \tan B)$$

$$\therefore \tan 70^\circ - \tan 20^\circ = \tan(70^\circ - 20^\circ)(1 + \tan 70^\circ \tan 20^\circ)$$

$$= \tan 50^\circ (1 + \tan 70^\circ \cot 70^\circ)$$

$$= 2 \tan 50^\circ$$

$$[\because 1 + \tan 70^\circ \cot 70^\circ = 2]$$

EXAMPLE 36 If $\tan(\alpha + \theta) = n \tan(\alpha - \theta)$, show that: $(n + 1) \sin 2\theta = (n - 1) \sin 2\alpha$.

SOLUTION We have,

$$\tan(\alpha + \theta) = n \tan(\alpha - \theta)$$

$$\Rightarrow \frac{\tan(\alpha + \theta)}{\tan(\alpha - \theta)} = \frac{n}{1}$$

$$\Rightarrow \frac{\tan(\alpha + \theta) + \tan(\alpha - \theta)}{\tan(\alpha + \theta) - \tan(\alpha - \theta)} = \frac{n + 1}{n - 1}$$

[Applying componendo-dividendo]

$$\Rightarrow \frac{\sin(\alpha + \theta) \cos(\alpha - \theta) + \cos(\alpha + \theta) \sin(\alpha - \theta)}{\sin(\alpha + \theta) \cos(\alpha - \theta) - \cos(\alpha + \theta) \sin(\alpha - \theta)} = \frac{n + 1}{n - 1}$$

$$\Rightarrow \frac{\sin(\alpha + \theta) + \alpha - \theta}{\sin(\alpha + \theta) - (\alpha - \theta)} = \frac{n + 1}{n - 1}$$

$$\Rightarrow \frac{\sin 2\alpha}{\sin 2\theta} = \frac{n + 1}{n - 1} \Rightarrow (n + 1) \sin 2\theta = (n - 1) \sin 2\alpha$$

EXAMPLE 37 Prove that: $\cot \theta \cot 2\theta + \cot 2\theta \cot 3\theta + 2 = \cot \theta (\cot \theta - \cot 3\theta)$

SOLUTION

$$\therefore \text{LHS} = \cot \theta \cot 2\theta + \cot 2\theta \cot 3\theta + 2$$

$$= (\cot \theta \cot 2\theta + 1) + (\cot 2\theta \cot 3\theta + 1)$$

$$= \left(\frac{\cos \theta \cos 2\theta}{\sin \theta \sin 2\theta} \right) + \left(\frac{\cos 2\theta \cos 3\theta}{\sin 2\theta \sin 3\theta} + 1 \right)$$

$$\begin{aligned}
 &= \left(\frac{\cos 2\theta \cos \theta + \sin 2\theta \sin \theta}{\sin \theta \sin 2\theta} \right) + \left(\frac{\cos 3\theta \cos 2\theta + \sin 3\theta \sin 2\theta}{\sin 2\theta \sin 3\theta} \right) \\
 &= \frac{\cos (2\theta - \theta)}{\sin \theta \sin 2\theta} + \frac{\cos (3\theta - 2\theta)}{\sin 2\theta \sin 3\theta} \\
 &= \frac{\cos \theta}{\sin \theta \sin 2\theta} + \frac{\cos \theta}{\sin 2\theta \sin 3\theta} \\
 &= \cos \theta \left\{ \frac{1}{\sin \theta \sin 2\theta} + \frac{1}{\sin 2\theta \sin 3\theta} \right\} \\
 &= \frac{\cos \theta}{\sin \theta} \left\{ \frac{\sin \theta}{\sin \theta \sin 2\theta} + \frac{\sin \theta}{\sin 2\theta \sin 3\theta} \right\} \\
 &= \cot \theta \left\{ \frac{\sin (2\theta - \theta)}{\sin \theta \sin 2\theta} + \frac{\sin (3\theta - 2\theta)}{\sin 2\theta \sin 3\theta} \right\} \\
 &= \cot \theta \left\{ \frac{\sin 2\theta \cos \theta - \cos 2\theta \sin \theta}{\sin \theta \sin 2\theta} + \frac{\sin 3\theta \cos 2\theta - \cos 3\theta \sin 2\theta}{\sin 2\theta \sin 3\theta} \right\} \\
 &= \cot \theta \{ \cot \theta - \cot 2\theta + \cot 2\theta - \cot 3\theta \} = \cot \theta (\cot \theta - \cot 3\theta) = \text{RHS}
 \end{aligned}$$

EXAMPLE 38 If $\tan (\pi \cos \theta) = \cot (\pi \sin \theta)$, prove that $\cos \left(\theta - \frac{\pi}{4} \right) = \pm \frac{1}{2\sqrt{2}}$.

SOLUTION We have,

$$\begin{aligned}
 &\tan (\pi \cos \theta) = \cot (\pi \sin \theta) \\
 \Rightarrow &\frac{\sin (\pi \cos \theta)}{\cos (\pi \cos \theta)} = \frac{\cos (\pi \sin \theta)}{\sin (\pi \sin \theta)} \\
 \Rightarrow &\sin (\pi \cos \theta) \sin (\pi \sin \theta) = \cos (\pi \sin \theta) \cos (\pi \cos \theta) \\
 \Rightarrow &\cos (\pi \cos \theta) \cos (\pi \sin \theta) - \sin (\pi \cos \theta) \sin (\pi \sin \theta) = 0 \\
 \Rightarrow &\cos (\pi \cos \theta + \pi \sin \theta) = 0 \\
 \Rightarrow &\pi \cos \theta + \pi \sin \theta = \pm \frac{\pi}{2} \quad \left[\because \cos \left(\pm \frac{\pi}{2} \right) = 0 \right] \\
 \Rightarrow &\cos \theta + \sin \theta = \pm \frac{1}{2} \\
 \Rightarrow &\frac{1}{\sqrt{2}} \cos \theta + \frac{1}{\sqrt{2}} \sin \theta = \pm \frac{1}{2\sqrt{2}} \quad [\text{Multiplying both sides by } 1/\sqrt{2}] \\
 \Rightarrow &\cos \theta \cos \frac{\pi}{4} + \sin \theta \sin \frac{\pi}{4} = \pm \frac{1}{2\sqrt{2}} \\
 \Rightarrow &\cos \left(\theta - \frac{\pi}{4} \right) = \pm \frac{1}{2\sqrt{2}}
 \end{aligned}$$

EXAMPLE 39 If $a \tan \alpha + b \tan \beta = (a+b) \tan \left(\frac{\alpha + \beta}{2} \right)$, where $\alpha \neq \beta$, prove that $a \cos \beta = b \cos \alpha$.

SOLUTION We have,

$$\begin{aligned}
 &a \tan \alpha + b \tan \beta = (a+b) \tan \left(\frac{\alpha + \beta}{2} \right) \\
 \Rightarrow &a \left\{ \tan \alpha - \tan \left(\frac{\alpha + \beta}{2} \right) \right\} = b \left\{ \tan \left(\frac{\alpha + \beta}{2} \right) - \tan \beta \right\}
 \end{aligned}$$

$$\Rightarrow \frac{a \sin \left(\alpha - \frac{\alpha + \beta}{2} \right)}{\cos \alpha \cos \left(\frac{\alpha + \beta}{2} \right)} = \frac{b \sin \left(\frac{\alpha + \beta}{2} - \beta \right)}{\cos \left(\frac{\alpha + \beta}{2} \right) \cos \beta}$$

$$\Rightarrow \frac{a \sin \left(\frac{\alpha - \beta}{2} \right)}{\cos \alpha} = \frac{b \sin \left(\frac{\alpha - \beta}{2} \right)}{\cos \beta}$$

$$\Rightarrow a \cos \beta = b \cos \alpha \quad \left[\because \alpha \neq \beta \therefore \sin \left(\frac{\alpha - \beta}{2} \right) \neq 0 \right]$$

EXAMPLE 40 If $\sin \alpha + \sin \beta = a$ and $\cos \alpha + \cos \beta = b$, show that

$$(i) \cos(\alpha + \beta) = \frac{b^2 - a^2}{b^2 + a^2} \quad (ii) \sin(\alpha + \beta) = \frac{2ab}{a^2 + b^2}$$

SOLUTION (i) We have,

$$\begin{aligned} b^2 + a^2 &= (\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 \\ \Rightarrow b^2 + a^2 &= (\cos^2 \alpha + \sin^2 \alpha) + (\cos^2 \beta + \sin^2 \beta) + 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\ \Rightarrow b^2 + a^2 &= 1 + 1 + 2 \cos(\alpha - \beta) = 2 + 2 \cos(\alpha - \beta) \quad \dots(i) \\ \text{and, } b^2 - a^2 &= (\cos \alpha + \cos \beta)^2 - (\sin \alpha + \sin \beta)^2 \\ \Rightarrow b^2 - a^2 &= \cos^2 \alpha + \cos^2 \beta - \sin^2 \alpha - \sin^2 \beta + 2(\cos \alpha \cos \beta - \sin \alpha \sin \beta) \\ \Rightarrow b^2 - a^2 &= (\cos^2 \alpha - \sin^2 \alpha) + (\cos^2 \beta - \sin^2 \beta) + 2 \cos(\alpha + \beta) \\ \Rightarrow b^2 - a^2 &= \cos(\alpha + \beta) \cos(\alpha - \beta) + \cos(\beta + \alpha) \cos(\beta - \alpha) + 2 \cos(\alpha + \beta) \\ \Rightarrow b^2 - a^2 &= 2 \cos(\alpha + \beta) \cos(\alpha - \beta) + 2 \cos(\alpha + \beta) \\ &\quad [\because \cos(\beta - \alpha) = \cos\{-(\alpha - \beta)\} = \cos(\alpha - \beta)] \\ \Rightarrow b^2 - a^2 &= \cos(\alpha + \beta) [2 \cos(\alpha - \beta) + 2] \\ \Rightarrow b^2 - a^2 &= \cos(\alpha + \beta) (b^2 + a^2) \quad [\text{Using (i)}] \end{aligned}$$

$$\text{Thus, } b^2 - a^2 = (b^2 + a^2) \cos(\alpha + \beta) \Rightarrow \cos(\alpha + \beta) = \frac{b^2 - a^2}{b^2 + a^2}$$

$$(ii) \sin(\alpha + \beta) = \sqrt{1 - \cos^2(\alpha + \beta)}$$

$$\Rightarrow \sin(\alpha + \beta) = \sqrt{1 - \left(\frac{b^2 - a^2}{b^2 + a^2} \right)^2} = \frac{\sqrt{4a^2b^2}}{\sqrt{(a^2 + b^2)^2}} = \frac{2ab}{a^2 + b^2}$$

EXAMPLE 41 If α and β are the solutions of the equation $a \tan \theta + b \sec \theta = c$, then show that $\tan(\alpha + \beta) = \frac{2ac}{a^2 - c^2}$ [NCERT EXEMPLAR]

SOLUTION We have,

$$\begin{aligned} a \tan \theta + b \sec \theta &= c \quad \dots(i) \\ \Rightarrow c - a \tan \theta &= b \sec \theta \\ \Rightarrow (c - a \tan \theta)^2 &= b^2 \sec^2 \theta \\ \Rightarrow c^2 + a^2 \tan^2 \theta - 2ac \tan \theta &= b^2 (1 + \tan^2 \theta) \\ \Rightarrow \tan^2 \theta (a^2 - b^2) - 2ac \tan \theta + (c^2 - b^2) &= 0 \quad \dots(ii) \end{aligned}$$

It is given that α & β are the solutions of (i). Therefore, $\tan \alpha$ and $\tan \beta$ are roots of equation (ii).

$$\therefore \tan \alpha + \tan \beta = \frac{2ac}{a^2 - b^2} \text{ and } \tan \alpha \tan \beta = \frac{c^2 - b^2}{a^2 - b^2}$$

$$\Rightarrow \tan (\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{\frac{2ac}{a^2 - b^2}}{1 - \frac{c^2 - b^2}{a^2 - b^2}} = \frac{2ac}{a^2 - c^2}$$

EXAMPLE 42 If α and β are the solutions of $a \cos \theta + b \sin \theta = c$, then show that

$$(i) \cos (\alpha + \beta) = \frac{a^2 - b^2}{a^2 + b^2} \quad (ii) \cos (\alpha - \beta) = \frac{2c^2 - (a^2 + b^2)}{a^2 + b^2}$$

[NCERT EXEMPLAR]

SOLUTION We have,

$$a \cos \theta + b \sin \theta = c \quad \dots(i)$$

$$\Rightarrow a \cos \theta = c - b \sin \theta$$

$$\Rightarrow a^2 \cos^2 \theta = (c - b \sin \theta)^2$$

$$\Rightarrow a^2 (1 - \sin^2 \theta) = c^2 - 2bc \sin \theta + b^2 \sin^2 \theta$$

$$\Rightarrow (a^2 + b^2) \sin^2 \theta - 2bc \sin \theta + (c^2 - a^2) = 0 \quad \dots(ii)$$

Since α , β are roots of equation (i). Therefore, $\sin \alpha$ and $\sin \beta$ are roots of equation (ii).

$$\therefore \sin \alpha \sin \beta = \frac{c^2 - a^2}{a^2 + b^2} \quad \dots(iii)$$

Again, $a \cos \theta + b \sin \theta = c$

$$\Rightarrow b \sin \theta = c - a \cos \theta$$

$$\Rightarrow b^2 \sin^2 \theta = (c - a \cos \theta)^2$$

$$\Rightarrow b^2 (1 - \cos^2 \theta) = (c - a \cos \theta)^2$$

$$\Rightarrow (a^2 + b^2) \cos^2 \theta - 2ac \cos \theta + c^2 - b^2 = 0 \quad \dots(iv)$$

It is given that α , β are the roots of equation (i). So, $\cos \alpha$, $\cos \beta$ are the roots of equation (iv).

$$\therefore \cos \alpha \cos \beta = \frac{c^2 - b^2}{a^2 + b^2} \quad \dots(v)$$

Now, $\cos (\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$

$$\Rightarrow \cos (\alpha + \beta) = \frac{c^2 - b^2}{a^2 + b^2} - \frac{c^2 - a^2}{a^2 + b^2} = \frac{a^2 - b^2}{a^2 + b^2}$$

and, $\cos (\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

$$\Rightarrow \cos (\alpha - \beta) = \frac{c^2 - b^2}{a^2 + b^2} + \frac{c^2 - a^2}{a^2 + b^2} = \frac{2c^2 - (a^2 + b^2)}{a^2 + b^2}$$

EXAMPLE 43 Prove that:

$$\frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x} = \frac{1}{2} (\tan 27x - \tan x)$$

SOLUTION We have,

$$\frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x}$$

$$\begin{aligned}
&= \frac{1}{2} \left\{ \frac{2 \sin x \cos x}{\cos 3x \cos x} + \frac{2 \sin 3x \cos 3x}{\cos 9x \cos 3x} + \frac{2 \sin 9x \cos 9x}{\cos 27x \cos 9x} \right\} \\
&= \frac{1}{2} \left\{ \frac{\sin (x+x)}{\cos 3x \cos x} + \frac{\sin (3x+3x)}{\cos 9x \cos 3x} + \frac{\sin (9x+9x)}{\cos 27x \cos 9x} \right\} \\
&= \frac{1}{2} \left\{ \frac{\sin 2x}{\cos 3x \cos x} + \frac{\sin 6x}{\cos 9x \cos 3x} + \frac{\sin 18x}{\cos 27x \cos 9x} \right\} \\
&= \frac{1}{2} \left\{ \frac{\sin (3x-x)}{\cos 3x \cos x} + \frac{\sin (9x-3x)}{\cos 9x \cos 3x} + \frac{\sin (27x-9x)}{\cos 27x \cos 9x} \right\} \\
&= \frac{1}{2} \left\{ \frac{\sin 3x \cos x - \cos 3x \sin x}{\cos 3x \cos x} + \frac{\sin 9x \cos 3x - \cos 9x \sin 3x}{\cos 9x \cos 3x} + \frac{\sin 27x \cos 9x - \cos 27x \sin 9x}{\cos 27x \cos 9x} \right\} \\
&= \frac{1}{2} \left\{ \frac{\sin 3x \cos x}{\cos 3x \cos x} - \frac{\cos 3x \sin x}{\cos 3x \cos x} + \frac{\sin 9x \cos 3x}{\cos 9x \cos 3x} - \frac{\cos 9x \sin 3x}{\cos 9x \cos 3x} + \frac{\sin 27x \cos 9x}{\cos 27x \cos 9x} - \frac{\cos 27x \sin 9x}{\cos 27x \cos 9x} \right\} \\
&= \frac{1}{2} \{ (\tan 3x - \tan x) + (\tan 9x - \tan 3x) + (\tan 27x - \tan 9x) \} = \frac{1}{2} (\tan 27x - \tan x)
\end{aligned}$$

EXERCISE 7.1**LEVEL-1**

- If $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$, where $0 < A, B < \frac{\pi}{2}$, find the values of the following:
 - $\sin (A+B)$
 - $\cos (A+B)$
 - $\sin (A-B)$
 - $\cos (A-B)$
- (a) If $\sin A = \frac{12}{13}$ and $\sin B = \frac{4}{5}$, where $\frac{\pi}{2} < A < \pi$ and $0 < B < \frac{\pi}{2}$, find the following:
 - $\sin (A+B)$
 - $\cos (A+B)$
 (b) If $\sin A = \frac{3}{5}$, $\cos B = -\frac{12}{13}$, where A and B both lie in second quadrant, find the value of $\sin (A+B)$.
- If $\cos A = -\frac{24}{25}$ and $\cos B = \frac{3}{5}$, where $\pi < A < \frac{3\pi}{2}$ and $\frac{3\pi}{2} < B < 2\pi$, find the following:
 - $\sin (A+B)$
 - $\cos (A+B)$
- If $\tan A = \frac{3}{4}$, $\cos B = \frac{9}{41}$, where $\pi < A < \frac{3\pi}{2}$ and $0 < B < \frac{\pi}{2}$, find $\tan (A+B)$.
- If $\sin A = \frac{1}{2}$, $\cos B = \frac{12}{13}$, where $\frac{\pi}{2} < A < \pi$ and $\frac{3\pi}{2} < B < 2\pi$, find $\tan (A-B)$.
- If $\sin A = \frac{1}{2}$, $\cos B = \frac{\sqrt{3}}{2}$, where $\frac{\pi}{2} < A < \pi$ and $0 < B < \frac{\pi}{2}$, find the following:
 - $\tan (A+B)$
 - $\tan (A-B)$
- Evaluate the following:
 - $\sin 78^\circ \cos 18^\circ - \cos 78^\circ \sin 18^\circ$
 - $\cos 47^\circ \cos 13^\circ - \sin 47^\circ \sin 13^\circ$
 - $\sin 36^\circ \cos 9^\circ + \cos 36^\circ \sin 9^\circ$
 - $\cos 80^\circ \cos 20^\circ + \sin 80^\circ \sin 20^\circ$
- If $\cos A = -\frac{12}{13}$ and $\cot B = \frac{24}{7}$, where A lies in the second quadrant and B in the third quadrant, find the values of the following:
 - $\sin (A+B)$
 - $\cos (A+B)$
 - $\tan (A+B)$
- Prove that: $\cos 105^\circ + \cos 15^\circ = \sin 75^\circ - \sin 15^\circ$
- Prove that: $\frac{\tan A + \tan B}{\tan A - \tan B} = \frac{\sin (A+B)}{\sin (A-B)}$

11. Prove that:

$$(i) \frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ} = \tan 56^\circ$$

$$(ii) \frac{\cos 9^\circ + \sin 9^\circ}{\cos 9^\circ - \sin 9^\circ} = \tan 54^\circ$$

$$(iii) \frac{\cos 8^\circ - \sin 8^\circ}{\cos 8^\circ + \sin 8^\circ} = \tan 37^\circ$$

12. Prove that:

$$(i) \sin (60^\circ - \theta) \cos (30^\circ + \theta) + \cos (60^\circ - \theta) \sin (30^\circ + \theta) = 1$$

$$(ii) \sin \left(\frac{4\pi}{9} + 7 \right) \cos \left(\frac{\pi}{9} + 7 \right) - \cos \left(\frac{4\pi}{9} + 7 \right) \sin \left(\frac{\pi}{9} + 7 \right) = \frac{\sqrt{3}}{2}$$

$$(iii) \sin \left(\frac{3\pi}{8} - 5 \right) \cos \left(\frac{\pi}{8} + 5 \right) + \cos \left(\frac{3\pi}{8} - 5 \right) \sin \left(\frac{\pi}{8} + 5 \right) = 1$$

$$13. \text{ Prove that: } \frac{\tan 69^\circ + \tan 66^\circ}{1 - \tan 69^\circ \tan 66^\circ} = -1$$

$$14. (i) \text{ If } \tan A = \frac{5}{6} \text{ and } \tan B = \frac{1}{11}, \text{ prove that } A + B = \frac{\pi}{4}$$

$$(ii) \text{ If } \tan A = \frac{m}{m-1} \text{ and } \tan B = \frac{1}{2m-1}, \text{ then prove that } A - B = \frac{\pi}{4}$$

15. Prove that:

$$(i) \cos^2 45^\circ - \sin^2 15^\circ = \frac{\sqrt{3}}{4}$$

$$(ii) \sin^2 (n+1) A - \sin^2 nA = \sin (2n+1) A \sin A$$

16. Prove that:

$$(i) \frac{\sin (A+B) + \sin (A-B)}{\cos (A+B) + \cos (A-B)} = \tan A$$

$$(ii) \frac{\sin (A-B)}{\cos A \cos B} + \frac{\sin (B-C)}{\cos B \cos C} + \frac{\sin (C-A)}{\cos C \cos A} = 0$$

$$(iii) \frac{\sin (A-B)}{\sin A \sin B} + \frac{\sin (B-C)}{\sin B \sin C} + \frac{\sin (C-A)}{\sin C \sin A} = 0$$

$$(iv) \sin^2 B = \sin^2 A + \sin^2 (A-B) - 2 \sin A \cos B \sin (A-B)$$

$$(v) \cos^2 A + \cos^2 B - 2 \cos A \cos B \cos (A+B) = \sin^2 (A+B)$$

$$(vi) \frac{\tan (A+B)}{\cot (A-B)} = \frac{\tan^2 A - \tan^2 B}{1 - \tan^2 A \tan^2 B}$$

17. Prove that:

$$(i) \tan 8\theta - \tan 6\theta - \tan 2\theta = \tan 8\theta \tan 6\theta \tan 2\theta$$

$$(ii) \tan 15^\circ + \tan 30^\circ + \tan 15^\circ \tan 30^\circ = 1$$

$$(iii) \tan 36^\circ + \tan 9^\circ + \tan 36^\circ \tan 9^\circ = 1$$

$$(iv) \tan 13\theta - \tan 9\theta - \tan 4\theta = \tan 13\theta \tan 9\theta \tan 4\theta$$

$$18. \text{ Prove that: } \frac{\tan^2 2\theta - \tan^2 \theta}{1 - \tan^2 2\theta \tan^2 \theta} = \tan 3\theta \tan \theta$$

$$19. \text{ If } \frac{\sin (x+y)}{\sin (x-y)} = \frac{a+b}{a-b}, \text{ show that } \frac{\tan x}{\tan y} = \frac{a}{b}.$$

$$20. \text{ If } \tan A = x \tan B, \text{ prove that } \frac{\sin (A-B)}{\sin (A+B)} = \frac{x-1}{x+1}$$

[NCERT EXAMPLAR]

21. If $\tan(A+B) = x$ and $\tan(A-B) = y$, find the values of $\tan 2A$ and $\tan 2B$.
 22. If $\cos A + \sin B = m$ and $\sin A + \cos B = n$, prove that $2 \sin(A+B) = m^2 + n^2 - 2$.

LEVEL-2

23. If $\tan A + \tan B = a$ and $\cot A + \cot B = b$, prove that: $\cot(A+B) = \frac{1}{a} - \frac{1}{b}$.
 24. If θ lies in the first quadrant and $\cos \theta = \frac{8}{17}$, then prove that

$$\cos\left(\frac{\pi}{6} + \theta\right) + \cos\left(\frac{\pi}{4} - \theta\right) + \cos\left(\frac{2\pi}{3} - \theta\right) = \left(\frac{\sqrt{3}-1}{2} + \frac{1}{\sqrt{2}}\right) \frac{23}{17}$$
 [NCERT EXEMPLAR]
 25. If $\tan x + \tan\left(x + \frac{\pi}{3}\right) + \tan\left(x + \frac{2\pi}{3}\right) = 3$, then prove that $\frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x} = 1$.
 26. If $\sin(\alpha + \beta) = 1$ and $\sin(\alpha - \beta) = \frac{1}{2}$, where $0 \leq \alpha, \beta \leq \frac{\pi}{2}$, then find the values of $\tan(\alpha + 2\beta)$ and $\tan(2\alpha + \beta)$.
 27. If α, β are two different values of θ lying between 0 and 2π which satisfy the equation $6 \cos \theta + 8 \sin \theta = 9$, find the value of $\sin(\alpha + \beta)$.
 28. If $\sin \alpha + \sin \beta = a$ and $\cos \alpha + \cos \beta = b$, show that
 (i) $\sin(\alpha + \beta) = \frac{2ab}{a^2 + b^2}$ (ii) $\cos(\alpha + \beta) = \frac{b^2 - a^2}{b^2 + a^2}$
 29. Prove that :
 (i) $\frac{1}{\sin(x-a) \sin(x-b)} = \frac{\cot(x-a) - \cot(x-b)}{\sin(a-b)}$
 (ii) $\frac{1}{\sin(x-a) \cos(x-b)} = \frac{\cot(x-a) + \tan(x-b)}{\cos(a-b)}$
 (iii) $\frac{1}{\cos(x-a) \cos(x-b)} = \frac{\tan(x-b) - \tan(x-a)}{\sin(a-b)}$
 30. If $\sin \alpha \sin \beta - \cos \alpha \cos \beta + 1 = 0$, prove that $1 + \cot \alpha \tan \beta = 0$.
 31. If $\tan \alpha = x + 1$, $\tan \beta = x - 1$, show that $2 \cot(\alpha - \beta) = x^2$.
 32. If angle θ is divided into two parts such that the tangents of one part is λ times the tangent of other, and ϕ is their difference, then show that $\sin \theta = \frac{\lambda + 1}{\lambda - 1} \sin \phi$.

[NCERT EXEMPLAR]

33. If $\tan \theta = \frac{\sin \alpha - \cos \alpha}{\sin \alpha + \cos \alpha}$, then show that $\sin \alpha + \cos \alpha = \sqrt{2} \cos \theta$

[NCERT EXEMPLAR]

ANSWERS

- | | | | |
|----------------------------|-----------------------|------------------------|--|
| 1. (i) $\frac{56}{65}$ | (ii) $\frac{-33}{65}$ | (iii) $\frac{-16}{65}$ | (iv) $\frac{63}{65}$ |
| 2. (a) (i) $\frac{16}{65}$ | (ii) $\frac{-63}{65}$ | (b) $\frac{-56}{65}$ | |
| 3. (i) $\frac{3}{5}$ | (ii) $\frac{-4}{5}$ | 4. $\frac{-187}{84}$ | 5. $\frac{5\sqrt{3}-12}{5+12\sqrt{3}}$ |
| 6. (i) 0 | (ii) $-\sqrt{3}$ | | |

7. (i) $\frac{\sqrt{3}}{2}$ (ii) $\frac{1}{2}$ (iii) $\frac{1}{\sqrt{2}}$ (iv) $\frac{1}{2}$
 8. (i) $\frac{-36}{325}$ (ii) $\frac{323}{325}$ (iii) $-\frac{36}{325}$ 18. $\frac{x+y}{1-xy}, \frac{x-y}{1+xy}$
 26. $-\sqrt{3}, -\frac{1}{\sqrt{3}}$ 27. $\frac{24}{25}$

HINTS TO SELECTED PROBLEMS

9. LHS = $\cos(90^\circ + 15^\circ) + \cos(90^\circ - 75^\circ) = -\sin 15^\circ + \sin 75^\circ = \text{RHS}$
 $\cos 11^\circ + \sin 11^\circ$
 11. LHS = $\frac{\cos 11^\circ}{\cos 11^\circ - \sin 11^\circ} = \frac{1 + \tan 11^\circ}{1 - \tan 11^\circ} = \frac{\tan 45^\circ + \tan 11^\circ}{1 - \tan 45^\circ \tan 11^\circ} = \tan(45^\circ + 11^\circ) = \tan 56^\circ = \text{RHS}$
 $\cos 11^\circ$
 29. (i) We have,

$$\frac{1}{\sin(x-a)\sin(x-b)} = \frac{1}{\sin(a-b)} \left\{ \frac{\sin(a-b)}{\sin(x-a)\sin(x-b)} \right\}$$

$$= \frac{1}{\sin(a-b)} \left[\frac{\sin\{(x-b)-(x-a)\}}{\sin(x-a)\sin(x-b)} \right]$$

$$= \frac{1}{\sin(a-b)} \left[\frac{\sin(x-b)\cos(x-a) - \cos(x-b)\sin(x-a)}{\sin(x-a)\sin(x-b)} \right]$$

$$= \frac{1}{\sin(a-b)} \left[\frac{\sin(x-b)\cos(x-a)}{\sin(x-a)\sin(x-b)} - \frac{\cos(x-b)\sin(x-a)}{\sin(x-a)\sin(x-b)} \right]$$

$$= \frac{1}{\sin(a-b)} [\cot(x-a) - \cot(x-b)]$$

Similarly, we can prove other two parts.

33. $\tan \theta = \frac{\sin \alpha - \cos \alpha}{\sin \alpha + \cos \alpha}$

$$\tan \theta = \frac{\frac{\sin \alpha - \cos \alpha}{\cos \alpha}}{\frac{\sin \alpha + \cos \alpha}{\cos \alpha}} \quad [\text{Dividing numerator and denominator by } \cos \alpha]$$

$$\Rightarrow \tan \theta = \frac{\tan \alpha - 1}{\tan \alpha + 1}$$

$$\Rightarrow \tan \theta = \tan \left(\alpha - \frac{\pi}{4} \right) \Rightarrow \theta = \alpha - \frac{\pi}{4} \Rightarrow \alpha = \theta + \frac{\pi}{4}$$

$$\therefore \sin \alpha + \cos \alpha = \sin \left(\theta + \frac{\pi}{4} \right) + \cos \left(\theta + \frac{\pi}{4} \right)$$

$$= \frac{1}{\sqrt{2}} (\sin \theta + \cos \theta) + \frac{1}{\sqrt{2}} (\cos \theta - \sin \theta) = \sqrt{2} \cos \theta$$

7.4 MAXIMUM AND MINIMUM VALUES OF TRIGONOMETRICAL EXPRESSIONS

We have learnt that for those values of θ for which trigonometrical functions are defined, we have

$$-1 \leq \sin \theta \leq 1, -1 \leq \cos \theta \leq 1, -\infty < \tan \theta < \infty \text{ cosec } \theta \geq 1 \text{ or } \text{cosec } \theta \leq -1, \\ \sec \theta \geq 1 \text{ or } \sec \theta \leq -1 \text{ and, } -\infty < \cot \theta < \infty$$

In this section, we will find the maximum and minimum values of trigonometrical expressions of the form $a \cos \theta + b \sin \theta$ for varying values of θ .

Let $f(\theta) = a \cos \theta + b \sin \theta$ Further, let $a = r \sin \alpha$ and $b = r \cos \alpha$. Then,

$$a^2 + b^2 = r^2 \sin^2 \alpha + r^2 \cos^2 \alpha \text{ and, } \frac{a}{b} = \frac{r \sin \alpha}{r \cos \alpha}$$

$$\Rightarrow a^2 + b^2 = r^2 (\sin^2 \alpha + \cos^2 \alpha) \text{ and, } \frac{a}{b} = \tan \alpha$$

$$\Rightarrow r = \sqrt{a^2 + b^2} \text{ and, } \tan \alpha = \frac{a}{b}$$

Substituting the values of a and b in $f(\theta)$, we obtain

$$f(\theta) = r \sin \alpha \cos \theta + r \cos \alpha \sin \theta = r (\sin \alpha \cos \theta + \cos \alpha \sin \theta) = r \sin (\alpha + \theta)$$

We know that

$$-1 \leq \sin (\alpha + \theta) \leq 1 \text{ for all } \theta$$

$$\Rightarrow -r \leq r \sin (\alpha + \theta) \leq r \text{ for all } \theta$$

[Multiplying throughout by r]

$$\Rightarrow -\sqrt{a^2 + b^2} \leq f(\theta) \leq \sqrt{a^2 + b^2} \text{ for all } \theta$$

$$\Rightarrow -\sqrt{a^2 + b^2} \leq a \cos \theta + b \sin \theta \leq \sqrt{a^2 + b^2} \text{ for all } \theta$$

Hence, maximum and minimum values of $a \cos \theta + b \sin \theta$ are $\sqrt{a^2 + b^2}$ and $-\sqrt{a^2 + b^2}$ respectively.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Find the maximum and minimum values of $7 \cos \theta + 24 \sin \theta$.

SOLUTION We know that the maximum and minimum values of $a \cos \theta + b \sin \theta$ are $\sqrt{a^2 + b^2}$ and $-\sqrt{a^2 + b^2}$ respectively. Hence, the maximum and minimum values of $7 \cos \theta + 24 \sin \theta$ are $\sqrt{7^2 + 24^2} = 25$ and $-\sqrt{7^2 + 24^2} = -25$ respectively.

EXAMPLE 2 Find the maximum and minimum values of the following expressions:

(i) $3 \cos \theta + 5 \sin \left(\theta - \frac{\pi}{6} \right)$

(ii) $4 \sin \theta - 3 \cos \theta + 7$

SOLUTION (i) Let $f(\theta) = 3 \cos \theta + 5 \sin \left(\theta - \frac{\pi}{6} \right)$. Then,

$$f(\theta) = 3 \cos \theta + 5 \left\{ \sin \theta \cos \frac{\pi}{6} - \cos \theta \sin \frac{\pi}{6} \right\} = 3 \cos \theta + \frac{5\sqrt{3}}{2} \sin \theta - \frac{5}{2} \cos \theta$$

$$\Rightarrow f(\theta) = \frac{1}{2} \cos \theta + \frac{5\sqrt{3}}{2} \sin \theta$$

$$\therefore -\sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \leq \frac{1}{2} \cos \theta + \frac{5\sqrt{3}}{2} \sin \theta \leq \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \text{ for all } \theta.$$

$$\therefore -\sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \leq f(\theta) \leq \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \text{ for all } \theta$$

$$\Rightarrow -\sqrt{\frac{1}{4} + \frac{75}{4}} \leq 3 \cos \theta + 5 \sin \left(\theta - \frac{\pi}{6} \right) \leq \sqrt{\frac{1}{4} + \frac{75}{4}} \text{ for all } \theta$$

$$\Rightarrow -\sqrt{19} \leq 3 \cos \theta + 5 \sin \left(\theta - \frac{\pi}{6} \right) \leq \sqrt{19} \text{ for all } \theta$$

Hence, $-\sqrt{19}$ and $\sqrt{19}$ are respectively the minimum and the maximum values of $3 \cos \theta + 5 \sin \left(\theta - \frac{\pi}{6} \right)$.

(ii) Let $f(\theta) = 4 \sin \theta - 3 \cos \theta + 7$

We know that

$$\begin{aligned} -\sqrt{4^2 + (-3)^2} &\leq 4 \sin \theta - 3 \cos \theta \leq \sqrt{4^2 + (-3)^2} \text{ for all } \theta \\ \Rightarrow -5 &\leq 4 \sin \theta - 3 \cos \theta \leq 5 \text{ for all } \theta \\ \Rightarrow -5 + 7 &\leq 4 \sin \theta - 3 \cos \theta + 7 \leq 5 + 7 \text{ for all } \theta \\ \Rightarrow 2 &\leq f(\theta) \leq 12 \text{ for all } \theta \end{aligned}$$

Hence, minimum and maximum values of $4 \sin \theta - 3 \cos \theta + 7$ are 2 and 12 respectively.

EXAMPLE 3 Prove that $5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3$ lies between -4 and 10 .

SOLUTION Let $f(\theta) = 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3$. Then,

$$\begin{aligned} f(\theta) &= 5 \cos \theta + 3 \left(\cos \theta \cos \frac{\pi}{3} - \sin \theta \sin \frac{\pi}{3} \right) + 3 = 5 \cos \theta + \frac{3}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta + 3 \\ \Rightarrow f(\theta) &= \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta + 3 \quad \dots(i) \\ \therefore -\sqrt{\left(\frac{13}{2} \right)^2 + \left(\frac{3\sqrt{3}}{2} \right)^2} &\leq \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta \leq \sqrt{\left(\frac{13}{2} \right)^2 + \left(\frac{3\sqrt{3}}{2} \right)^2} \\ \Rightarrow -7 &\leq \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta \leq 7 \text{ for all } \theta \\ \Rightarrow -7 + 3 &\leq \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta + 3 \leq 7 + 3 \text{ for all } \theta \\ \Rightarrow -4 &\leq 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3 \leq 10 \text{ for all } \theta \quad [\text{Using (i)}] \end{aligned}$$

EXAMPLE 4 Find a and b such that the following inequality holds good for all θ :

$$a \leq 3 \cos \theta + 5 \sin \left(\theta - \frac{\pi}{6} \right) \leq b$$

SOLUTION Let $f(\theta) = 3 \cos \theta + 5 \sin \left(\theta - \frac{\pi}{6} \right)$. Then,

$$\begin{aligned} f(\theta) &= 3 \cos \theta + 5 \left(\sin \theta \cos \frac{\pi}{6} - \cos \theta \sin \frac{\pi}{6} \right) = 3 \cos \theta + 5 \left(\frac{\sqrt{3}}{2} \sin \theta - \frac{1}{2} \cos \theta \right) \\ \Rightarrow f(\theta) &= \frac{1}{2} \cos \theta + \frac{5\sqrt{3}}{2} \sin \theta \end{aligned}$$

We have,

$$\begin{aligned} \therefore \sqrt{\left(\frac{1}{2} \right)^2 + \left(\frac{5\sqrt{3}}{2} \right)^2} &\leq \frac{1}{2} \cos \theta + \frac{5\sqrt{3}}{2} \sin \theta \leq \sqrt{\left(\frac{1}{2} \right)^2 + \left(\frac{5\sqrt{3}}{2} \right)^2} \text{ for all } \theta \\ \Rightarrow -\sqrt{19} &\leq 3 \cos \theta + 5 \sin \left(\theta - \frac{\pi}{6} \right) \leq \sqrt{19} \text{ for all } \theta \end{aligned}$$

Hence, $a = -\sqrt{19}$ and $b = \sqrt{19}$

7.5 TO EXPRESS $a \cos \theta + b \sin \theta$ IN THE FORM $r \sin (\theta \pm \alpha)$ OR $r \cos (\theta \pm \alpha)$

Sometimes we need to express trigonometrical expressions of the form $a \cos \theta + b \sin \theta$ in terms of sine or cosine of single term. We may use the following algorithm to do so.

ALGORITHM

STEP I Multiply and divide $f(\theta) = a \cos \theta + b \sin \theta$ by $\sqrt{a^2 + b^2}$ to get

$$f(\theta) = \sqrt{a^2 + b^2} \left\{ \frac{a}{\sqrt{a^2 + b^2}} \cos \theta + \frac{b}{\sqrt{a^2 + b^2}} \sin \theta \right\}$$

STEP II In order to express $f(\theta)$ in terms of sine of some term, replace $\frac{a}{\sqrt{a^2 + b^2}}$ i.e. coefficient of

$\cos \theta$ by $\sin \alpha$ and $\frac{b}{\sqrt{a^2 + b^2}}$ i.e. coefficient of $\sin \theta$ by $\cos \alpha$. This gives the following :

$$f(\theta) = \sqrt{a^2 + b^2} \{ \sin \alpha \cos \theta + \cos \alpha \sin \theta \} = \sqrt{a^2 + b^2} \sin(\theta + \alpha)$$

To express $f(\theta)$ in terms of cosine of some term, replace coefficient of $\cos \theta$ i.e. $\frac{a}{\sqrt{a^2 + b^2}}$

by $\cos \alpha$ and coefficient of $\sin \theta$ i.e. $\frac{b}{\sqrt{a^2 + b^2}}$ by $\sin \alpha$. This gives the following:

$$f(\theta) = \sqrt{a^2 + b^2} \{ \cos \alpha \cos \theta + \sin \alpha \sin \theta \} = \sqrt{a^2 + b^2} \cos(\theta - \alpha).$$

ILLUSTRATIVE EXAMPLES**LEVEL-1**

EXAMPLE 1 Reduce $\sqrt{3} \sin \theta + \cos \theta$ as a single term consisting (i) sine only (ii) cosine only.

SOLUTION Let $f(\theta) = \sqrt{3} \sin \theta + \cos \theta$. Then,

$$f(\theta) = \sqrt{3} \sin \theta + \cos \theta$$

$$\Rightarrow f(\theta) = 2 \left\{ \frac{\sqrt{3}}{2} \sin \theta + \frac{1}{2} \cos \theta \right\} \quad \left[\text{Multiplying and dividing by } \sqrt{(\sqrt{3})^2 + 1^2} \text{ i.e. by } 2 \right]$$

$$\Rightarrow f(\theta) = 2 \left\{ \sin \theta \cos \frac{\pi}{6} + \cos \theta \sin \frac{\pi}{6} \right\} \quad \left[\because \frac{\sqrt{3}}{2} = \cos \frac{\pi}{6} \text{ and } \frac{1}{2} = \sin \frac{\pi}{6} \right]$$

$$\Rightarrow f(\theta) = 2 \sin \left(\theta + \frac{\pi}{6} \right)$$

Again,

$$f(\theta) = 2 \left\{ \frac{\sqrt{3}}{2} \sin \theta + \frac{1}{2} \cos \theta \right\} = 2 \left\{ \sin \frac{\pi}{3} \sin \theta + \cos \frac{\pi}{3} \cos \theta \right\}$$

$$\Rightarrow f(\theta) = 2 \left\{ \cos \theta \cos \frac{\pi}{3} + \sin \theta \sin \frac{\pi}{3} \right\} = 2 \cos \left(\theta - \frac{\pi}{3} \right).$$

EXAMPLE 2 Express $3 \cos \theta - 4 \sin \theta$ as sines and cosines of a single expression.

SOLUTION Let $f(\theta) = 3 \cos \theta - 4 \sin \theta$. Multiplying and dividing by $\sqrt{3^2 + (-4)^2}$ i.e. by 5, we get

$$f(\theta) = \sqrt{3^2 + (-4)^2} \left\{ \frac{3}{\sqrt{3^2 + (-4)^2}} \cos \theta - \frac{4}{\sqrt{3^2 + (-4)^2}} \sin \theta \right\}$$

$$\Rightarrow f(\theta) = 5 \left(\frac{3}{5} \cos \theta - \frac{4}{5} \sin \theta \right)$$

$$\Rightarrow f(\theta) = 5 (\sin \alpha \cos \theta - \cos \alpha \sin \theta), \text{ where } \sin \alpha = \frac{3}{5} \text{ and } \cos \alpha = \frac{4}{5}$$

$$\Rightarrow f(\theta) = 5 \sin(\alpha - \theta), \text{ where } \tan \alpha = \frac{3}{4}$$

Again,

$$f(\theta) = 5 \left(\frac{3}{5} \cos \theta - \frac{4}{5} \sin \theta \right)$$

$$\Rightarrow f(\theta) = 5(\cos \alpha \cos \theta - \sin \alpha \sin \theta), \text{ where } \cos \alpha = \frac{3}{5} \text{ and } \sin \alpha = \frac{4}{5}$$

$$\Rightarrow f(\theta) = 5 \cos(\alpha + \theta), \text{ where } \tan \alpha = \frac{4}{3}$$

EXAMPLE 3 Find the sign of the expression $\sin 100^\circ + \cos 100^\circ$.

SOLUTION We have,

$$\begin{aligned} \sin 100^\circ + \cos 100^\circ &= \sqrt{2} \left(\frac{1}{\sqrt{2}} \sin 100^\circ + \frac{1}{\sqrt{2}} \cos 100^\circ \right) \\ &= \sqrt{2} (\cos 45^\circ \sin 100^\circ + \sin 45^\circ \cos 100^\circ) \\ &= \sqrt{2} \sin(100^\circ + 45^\circ) \\ &= \sqrt{2} \sin 145^\circ, \text{ which is a positive real number. } [\because \sin 145^\circ \text{ is positive}] \end{aligned}$$

EXERCISE 7.2

LEVEL-1

- Find the maximum and minimum values of each of the following trigonometrical expressions:
 - $12 \sin \theta - 5 \cos \theta$
 - $12 \cos \theta + 5 \sin \theta + 4$
 - $5 \cos \theta + 3 \sin \left(\frac{\pi}{6} - \theta \right) + 4$
 - $\sin \theta - \cos \theta + 1$
- Reduce each of the following expressions to the sine and cosine of a single expression:
 - $\sqrt{3} \sin \theta - \cos \theta$
 - $\cos \theta - \sin \theta$
 - $24 \cos \theta + 7 \sin \theta$
- Show that $\sin 100^\circ - \sin 10^\circ$ is positive.
- Prove that $(2\sqrt{3} + 3) \sin \theta + 2\sqrt{3} \cos \theta$ lies between $-(2\sqrt{3} + \sqrt{15})$ and $(2\sqrt{3} + \sqrt{15})$.

ANSWERS

- | | |
|---------------------|----------------|
| 1. Minimum | Maximum |
| (i) -13 | 13 |
| (ii) -9 | 17 |
| (iii) -3 | 11 |
| (iv) $1 - \sqrt{2}$ | $1 + \sqrt{2}$ |
- $2 \sin \left(\theta - \frac{\pi}{6} \right), -2 \cos \left(\frac{\pi}{3} + \theta \right)$
 - $\sqrt{2} \sin \left(\frac{\pi}{4} - \theta \right), \sqrt{2} \cos \left(\frac{\pi}{4} + \theta \right)$
 - $25 \sin(\alpha + \theta), \text{ where } \tan \alpha = \frac{24}{7}, 25 \cos(\theta - \alpha), \text{ where } \tan \alpha = \frac{7}{24}$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

- If $\alpha + \beta - \gamma = \pi$, and $\sin^2 \alpha + \sin^2 \beta - \sin^2 \gamma = \lambda \sin \alpha \sin \beta \cos \gamma$, then write the value of λ .
- If $x \cos \theta = y \cos \left(\theta + \frac{2\pi}{3} \right) = z \cos \left(\theta + \frac{4\pi}{3} \right)$, then write the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.
- Write the maximum and minimum values of $3 \cos x + 4 \sin x + 5$.
- Write the maximum value of $12 \sin \theta - 9 \sin^2 \theta$.
- If $12 \sin \theta - 9 \sin^2 \theta$ attains its maximum value at $\theta = \alpha$, then write the value of $\sin \alpha$.

6. Write the interval in which the values of $5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3$ lie.
7. If $\tan (A + B) = p$ and $\tan (A - B) = q$, then write the value of $\tan 2B$.
8. If $\frac{\cos (x - y)}{\cos (x + y)} = \frac{m}{n}$, then write the value of $\tan x \tan y$.
9. If $a = b \cos \frac{2\pi}{3} = c \cos \frac{4\pi}{3}$, then write the value of $ab + bc + ca$.
10. If $A + B = C$, then write the value of $\tan A \tan B \tan C$.
11. If $\sin \alpha - \sin \beta = a$ and $\cos \alpha + \cos \beta = b$, then write the value of $\cos (\alpha + \beta)$.
12. If $\tan \alpha = \frac{1}{1 + 2^{-x}}$ and $\tan \beta = \frac{1}{1 + 2^{x+1}}$, then write the value of $\alpha + \beta$ lying in the interval $(0, \pi/2)$.

ANSWERS

1. 2 2. 0 3. Maximum = 10, Minimum = 0 4. 4 5. $\frac{2}{3}$
6. $[-4, 10]$ 7. $\frac{p-q}{1+pq}$ 8. $\frac{m-n}{m+n}$ 9. 0 10. $\tan C - \tan A - \tan B$
11. $\frac{a^2 + b^2 - 2}{2}$ 12. $\frac{\pi}{4}$

MULTIPLE CHOICE QUATIONS (MCQs)

Mark the correct alternative in each of the following:

1. The value of $\sin^2 75^\circ - \sin^2 15^\circ$ is
 (a) $1/2$ (b) $\sqrt{3}/2$ (c) 1 (d) 0
2. If $A + B + C = 180^\circ$, then $\sec A (\cos B \cos C - \sin B \sin C)$ is equal to
 (a) 0 (b) -1 (c) 1 (d) none of these
3. $\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ$ is equal to
 (a) $\frac{\sqrt{3}}{4}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\sqrt{3}$ (d) 1
4. If $\tan A = \frac{a}{a+1}$ and $\tan B = \frac{1}{2a+1}$, then the value of $A + B$ is
 (a) 0 (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{4}$
5. If $3 \sin \theta + 4 \cos \theta = 5$, then $4 \sin \theta - 3 \cos \theta =$
 (a) 0 (b) 5 (c) 1 (d) none of these
6. If in a $\triangle ABC$, $\tan A + \tan B + \tan C = 6$, then $\cot A \cot B \cot C =$
 (a) 6 (b) 1 (c) $1/6$ (d) none of these
7. $\tan 3A - \tan 2A - \tan A$ is equal to
 (a) $\tan 3A \tan 2A \tan A$ (b) $-\tan 3A \tan 2A \tan A$
 (c) $\tan A \tan 2A - \tan 2A \tan 3A - \tan 3A \tan A$
 (d) none of these.
8. If $A + B + C = 180^\circ$, then $\frac{\tan A + \tan B + \tan C}{\tan A \tan B \tan C}$ is equal to
 (a) $\tan A \tan B \tan C$ (b) 0
 (c) 1 (d) none of these.

9. If $\cos P = \frac{1}{7}$ and $\cos Q = \frac{13}{14}$, where P and Q both are acute angles. Then, the value of $P - Q$ is
 (a) 30° (b) 60° (c) 45° (d) 75°
10. If $\cot(\alpha + \beta) = 0$, then $\sin(\alpha + 2\beta)$ is equal to
 (a) $\sin \alpha$ (b) $\cos 2\beta$ (c) $\cos \alpha$ (d) $\sin 2\alpha$
11. $\frac{\cos 10^\circ + \sin 10^\circ}{\cos 10^\circ - \sin 10^\circ}$ is equal to
 (a) $\tan 55^\circ$ (b) $\cot 55^\circ$ (c) $-\tan 35^\circ$ (d) $-\cot 35^\circ$
12. The value of $\cos^2\left(\frac{\pi}{6} + \theta\right) - \sin^2\left(\frac{\pi}{6} - \theta\right)$ is
 (a) $\frac{1}{2} \cos 2\theta$ (b) 0 (c) $-\frac{1}{2} \cos 2\theta$ (d) $\frac{1}{2}$
13. If $\tan \theta_1 \tan \theta_2 = k$, then $\frac{\cos(\theta_1 - \theta_2)}{\cos(\theta_1 + \theta_2)} =$
 (a) $\frac{1+k}{1-k}$ (b) $\frac{1-k}{1+k}$ (c) $\frac{k+1}{k-1}$ (d) $\frac{k-1}{k+1}$
14. If $\sin(\pi \cos \theta) = \cos(\pi \sin \theta)$, then $\sin 2\theta =$
 (a) $\pm \frac{3}{4}$ (b) $\pm \frac{4}{3}$ (c) $\pm \frac{1}{3}$ (d) none of these
15. If $\tan \theta = \frac{1}{2}$ and $\tan \phi = \frac{1}{3}$, then the value of $\theta + \phi$ is
 (a) $\frac{\pi}{6}$ (b) π (c) 0 (d) $\frac{\pi}{4}$
16. The value of $\cos(36^\circ - A) \cos(36^\circ + A) + \cos(54^\circ + A) \cos(54^\circ - A)$ is
 (a) $\sin 2A$ (b) $\cos 2A$ (c) $\cos 3A$ (d) $\sin 3A$
17. If $\tan(\pi/4 + \theta) + \tan(\pi/4 - \theta) = a$, then $\tan^2(\pi/4 + \theta) + \tan^2(\pi/4 - \theta) =$
 (a) $a^2 + 1$ (b) $a^2 + 2$ (c) $a^2 - 2$ (d) none of these
18. If $\tan(A - B) = 1$, $\sec(A + B) = \frac{2}{\sqrt{3}}$, then the smallest positive value of B is
 (a) $\frac{25\pi}{24}$ (b) $\frac{19\pi}{24}$ (c) $\frac{13\pi}{24}$ (d) $\frac{11\pi}{24}$
19. If $A - B = \pi/4$, then $(1 + \tan A)(1 - \tan B)$ is equal to
 (a) 2 (b) 1 (c) 0 (d) 3
20. The maximum value of $\sin^2(120^\circ + \theta) + \sin^2(120^\circ - \theta)$ is
 (a) $1/2$ (b) $3/2$ (c) $1/4$ (d) $3/4$
21. If $\cos(A - B) = \frac{3}{5}$ and $\tan A \tan B = 2$, then
 (a) $\cos A \cos B = \frac{1}{5}$ (b) $\cos A \cos B = -\frac{1}{5}$
 (c) $\sin A \sin B = -\frac{1}{5}$ (d) $\sin A \sin B = -\frac{1}{5}$
22. If $\tan 69^\circ + \tan 66^\circ - \tan 69^\circ \tan 66^\circ = 2k$, then $k =$
 (a) -1 (b) $1/2$ (c) $-1/2$ (d) none of these
23. If $\tan \alpha = \frac{x}{x+1}$ and $\tan \beta = \frac{1}{2x+1}$, then $\alpha + \beta$ is equal to
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$

ANSWERS

- | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (c) | 4. (d) | 5. (a) | 6. (c) | 7. (a) | 8. (c) |
| 9. (b) | 10. (a) | 11. (a) | 12. (a) | 13. (a) | 14. (a) | 15. (d) | 16. (b) |
| 17. (c) | 18. (b) | 19. (a) | 20. (b) | 21. (a) | 22. (c) | 23. (d) | |

SUMMARY

1. (i) $\sin(A + B) = \sin A \cos B + \cos A \sin B$
 (ii) $\sin(A - B) = \sin A \cos B - \cos A \sin B$
 (iii) $\cos(A + B) = \cos A \cos B - \sin A \sin B$
 (iv) $\cos(A - B) = \cos A \cos B + \sin A \sin B$
 (v) $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
 (vi) $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
 (vii) $\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B$
 (viii) $\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B$
2. (i) $\sin(A + B + C) = \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C$
 (ii) $\cos(A + B + C) = \cos A \cos B \cos C - \cos A \sin B \sin C - \sin A \cos B \sin C - \sin A \sin B \cos C$
 (iii) $\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$
3. (i) If $A + B = \pi$, then $\sin A = \sin B$, $\cos A = -\cos B$ and $\tan A = -\tan B$
 (ii) If $A + B = 2\pi$, then $\sin A = -\sin B$, $\cos A = \cos B$ and $\tan A = -\tan B$

TRANSFORMATION FORMULAE

8.1 INTRODUCTION

In this chapter, we will establish two sets of transformation formulae: One to transform the products of two sines or two cosines or one sine and one cosine into the sum or difference of two sines or two cosines and the other to convert the sum or difference of two sines or two cosines in the product of two sines or two cosines or one sine and one cosine. These two sets of formulae are of fundamental importance and one should have thorough acquaintance with these formulae.

8.2 FORMULAE TO TRANSFORM THE PRODUCT INTO SUM OR DIFFERENCE

In the previous chapter we have derived the following formulae:

$$\sin A \cos B + \cos A \sin B = \sin (A + B) \quad \dots(i)$$

$$\sin A \cos B - \cos A \sin B = \sin (A - B) \quad \dots(ii)$$

$$\cos A \cos B - \sin A \sin B = \cos (A + B) \quad \dots(iii)$$

$$\cos A \cos B + \sin A \sin B = \cos (A - B) \quad \dots(iv)$$

Adding (i) and (ii), we obtain

$$2 \sin A \cos B = \sin (A + B) + \sin (A - B)$$

Subtracting (ii) from (i), we get

$$2 \cos A \sin B = \sin (A + B) - \sin (A - B)$$

Adding (iii) and (iv), we get

$$2 \cos A \cos B = \cos (A + B) + \cos (A - B)$$

Subtracting (iii) from (iv), we get

$$2 \sin A \sin B = \cos (A - B) - \cos (A + B)$$

Thus, we obtain the following formulae :

$$(a) \ 2 \sin A \cos B = \sin (A + B) + \sin (A - B) \quad (b) \ 2 \cos A \sin B = \sin (A + B) - \sin (A - B)$$

$$(c) \ 2 \cos A \cos B = \cos (A + B) + \cos (A - B) \quad (d) \ 2 \sin A \sin B = \cos (A - B) - \cos (A + B)$$

These four formulae convert the product of two sines or two cosines or one sine and one cosine into the sum or difference of two sines or two cosines.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Convert each of the following products into the sum or difference of sines and cosines:

$$(i) \ 2 \sin 5\theta \cos \theta \quad (ii) \ 2 \cos 4\theta \cos 3\theta$$

$$(iii) \ 2 \sin 3\theta \sin \theta$$

$$(iv) \ \sin 75^\circ \cos 15^\circ \quad (v) \ \cos 75^\circ \cos 15^\circ$$

SOLUTION (i) Using $2 \sin A \cos B = \sin (A + B) + \sin (A - B)$, we obtain

$$2 \sin 5\theta \cos \theta = \sin (5\theta + \theta) + \sin (5\theta - \theta) = \sin 6\theta + \sin 4\theta$$

(ii) Using $2 \sin A \sin B = \cos (A - B) - \cos (A + B)$, we obtain

$$2 \cos 4\theta \cos 3\theta = \cos (4\theta + 3\theta) + \cos (4\theta - 3\theta) = \cos 7\theta + \cos \theta$$

(iii) Using $2 \sin A \sin B = \cos (A - B) - \cos (A + B)$, we obtain

$$2 \sin 3\theta \sin \theta = \cos(3\theta - \theta) - \cos(3\theta + \theta) = \cos 2\theta - \cos 4\theta$$

$$\begin{aligned} \text{(iv) } \sin 75^\circ \cos 15^\circ &= \frac{1}{2} (2 \sin 75^\circ \cos 15^\circ) \\ &= \frac{1}{2} \{\sin(75^\circ + 15^\circ) + \sin(75^\circ - 15^\circ)\} = \frac{1}{2} (\sin 90^\circ + \sin 60^\circ) \end{aligned}$$

$$\begin{aligned} \text{(v) } \cos 75^\circ \cos 15^\circ &= \frac{1}{2} (2 \cos 75^\circ \cos 15^\circ) \\ &= \frac{1}{2} \{\cos(75^\circ + 15^\circ) + \cos(75^\circ - 15^\circ)\} = \frac{1}{2} (\cos 90^\circ + \cos 60^\circ) \end{aligned}$$

EXAMPLE 2 Prove that: $2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0$

[NCERT]

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= 2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} \\ &= \cos \left(\frac{9\pi}{13} + \frac{\pi}{13} \right) + \cos \left(\frac{9\pi}{13} - \frac{\pi}{13} \right) + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} \\ &= \cos \frac{10\pi}{13} + \cos \frac{8\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} \\ &= \cos \left(\pi - \frac{3\pi}{13} \right) + \cos \left(\pi - \frac{5\pi}{13} \right) + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} \\ &= -\cos \frac{3\pi}{13} - \cos \frac{5\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0 = \text{RHS} \quad [\because \cos(\pi - \theta) = -\cos \theta] \end{aligned}$$

EXAMPLE 3 Prove that: $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

SOLUTION We have,

$$\begin{aligned} \Rightarrow \text{LHS} &= \cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ \\ \Rightarrow \text{LHS} &= \cos 60^\circ (\cos 20^\circ \cos 40^\circ) \cos 80^\circ \\ \Rightarrow \text{LHS} &= \frac{1}{2} \times \frac{1}{2} (2 \cos 20^\circ \cos 40^\circ) \cos 80^\circ \quad \left[\because \cos 60^\circ = \frac{1}{2} \right] \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left[\{\cos(40^\circ + 20^\circ) + \cos(40^\circ - 20^\circ)\} \cos 80^\circ \right] \\ &\quad [\because 2 \cos A \cos B = \cos(A + B) + \cos(A - B)] \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left\{ (\cos 60^\circ + \cos 20^\circ) \cos 80^\circ \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left\{ \left(\frac{1}{2} + \cos 20^\circ \right) \cos 80^\circ \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left\{ \frac{1}{2} \cos 80^\circ + \cos 80^\circ \cos 20^\circ \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{8} \left\{ \cos 80^\circ + 2 \cos 80^\circ \cos 20^\circ \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{8} \left[\cos 80^\circ + \left\{ \cos(80^\circ + 20^\circ) + \cos(80^\circ - 20^\circ) \right\} \right] \\ &\quad [\because 2 \cos A \cos B = \cos(A + B) + \cos(A - B)] \\ \Rightarrow \text{LHS} &= \frac{1}{8} \{ \cos 80^\circ + \cos 100^\circ + \cos 60^\circ \} \\ \Rightarrow \text{LHS} &= \frac{1}{8} \{ \cos 80^\circ + \cos(180^\circ - 80^\circ) + \cos 60^\circ \} \end{aligned}$$